

PRESENTATION OF BOREL SUBGROUPS OF KAC–MOODY GROUPS OVER LOCAL RINGS

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ABSTRACT. We obtain presentations of Borel subgroups of certain Chevalley groups and Kac–Moody groups of simply laced type. More precisely, we show that Borel subgroups of simply laced Chevalley groups over a commutative, unital ring (satisfying a minor technical condition), are the amalgamated product of their rank 2 subgroups, generalising a result of Tits over fields (different from $\mathbb{F}_2$). We moreover show that Borel subgroups of Kac–Moody groups over local Bézout domains also have a presentation as an amalgamated product of their rank 2 subgroups as soon as this holds for their rank 3 subgroups. In particular we deduce that Borel subgroups of simply laced Kac–Moody groups over a local Bézout domain (with residue field different from $\mathbb{F}_2$) have such a presentation if and only if they are of 3-spherical type.

1. INTRODUCTION

In this article, we show that the Borel subgroups of simply laced Kac–Moody groups over a local Bézout domain (satisfying the minor condition that its residue field is different from $\mathbb{F}_2$) have a presentation as the amalgamated product of a certain finite system of subgroups if and only if the Kac–Moody group is of 3-spherical type.

A (minimal) Kac–Moody group $\mathfrak{G}_A(R)$ over a ring R is constructed from a generalised Cartan matrix (GCM) A . More precisely, associated to any GCM A , there is a Coxeter group $\mathcal{W}$ (the Weyl group of A), and the group $\mathfrak{G}_A(R)$ is defined by a Steinberg-like presentation in terms of generators $x_\alpha(r)$, for all $r \in R$ and all roots α in the root system Φ of the Coxeter complex of $\mathcal{W}$. If $\mathcal{W}$ is infinite (or, equivalently, A is not a Cartan matrix), there are infinitely many roots α , and hence $\mathfrak{G}_A(R)$ is defined by a presentation in terms of infinitely many generators (and infinitely many relations).

Inside $\mathfrak{G}_A(R)$, there is a Borel subgroup B_R^+ , consisting of a “maximal unipotent” subgroup U_R^+ generated by the elements $x_\alpha(r)$ for all $\alpha \in \Phi$ positive, as well as a torus T_R .

In the literature, the question of obtaining tractable presentations of the groups U_R^+ (and hence B_R^+) has mostly been considered over fields. In [Tit86, Proposition 7] (see also Proposition 5.1 below), Tits proved that over most rings R , the maximal unipotent subgroup U_R^+ of a Chevalley group (i.e. a Kac–Moody group of *spherical* type, meaning that A is a Cartan matrix) is the amalgamated product of its rank 2 subgroups as soon as this holds for all of its rank 3 subgroups. He moreover showed that when $\mathbb{K}$ is a field different from $\mathbb{F}_2$, the group $U_{\mathbb{K}}^+$ of type A_3 (i.e. the group of upper triangular unipotent matrices in $\mathrm{SL}_4(\mathbb{K})$) has such a presentation.

Later, using geometric arguments, Abramenko obtained in [Abr96, Proposition 11] the same result for the maximal unipotent subgroup $U_{\mathbb{K}}^+$ of Chevalley groups of type B_3 and C_3 over fields $\mathbb{K}$, provided that $\mathbb{K}$ has size at least 16. This bound was improved to 5 in [HNVM16]. In combination with the reduction step in Tits’ work, this effectively settled the question of obtaining such presentations of the subgroup $U_{\mathbb{K}}^+$ of Chevalley groups over fields $\mathbb{K}$.

Over more general rings, the same question remains open. In this paper, we adapt Tits’ strategy in [Tit86] to prove the following. For any root α , denote by U_α the subgroup of U_R^+ generated

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by $x_\alpha(r)$ for all $r \in R$. Recall that a GCM $A = (A_{ij})_{i,j \in I}$ is called *simply laced* if $A_{ij}A_{ji} \leq 1$ for all distinct $i, j \in I$.

Theorem A. *Let $A = (A_{ij})_{i,j \in I}$ be a simply laced Cartan matrix. Let R be a commutative, unital ring such that 1_R is the sum of 2 invertible elements. Then $U_R^+ \leq \mathfrak{G}_A(R)$ is the amalgamated product of its subgroups $U_{i,j} := \langle U_{\alpha_i}, U_{\alpha_j} \rangle$, along the canonical inclusions $U_{\alpha_i} \hookrightarrow U_{i,j}$, for all $i, j \in I$.*

Over semi-local rings, the condition that 1_R is the sum of two invertible elements is equivalent to the ring having no quotient $\mathbb{F}_2$ (see Proposition 5.9 below).

In [DM07], similar results for Kac–Moody groups of non-spherical type over fields $\mathbb{K}$ are obtained. Using geometric methods rather than group theoretic ones as in Tits’ work, the authors again make a reduction step, showing that $U_{\mathbb{K}}^+$ has a presentation in terms of an amalgamated product of its rank 2 subgroups, if this is the case for each rank 3 subgroup. Using then the aforementioned results for the A_3 ([Tit86]), B_3 and C_3 cases ([Abr96] and [HNVM16]), this yields the desired presentation for $U_{\mathbb{K}}^+ \leq \mathfrak{G}_A(\mathbb{K})$, with A of 3-spherical type, and $\mathbb{K}$ a field of size at least 5. Note that a GCM $A = (A_{ij})_{i,j \in I}$ is called *3-spherical* if $A_J := (A_{ij})_{i,j \in J}$ is a Cartan matrix for all $J \subseteq I$ with $|J| = 3$.

It was moreover recently shown in [CM26, Theorem E] that A being of 3-spherical type is a necessary condition for $U_{\mathbb{K}}^+$ to have this presentation. These results together provide a reasonably good understanding of presentations of $U_{\mathbb{K}}^+$ in the Kac–Moody setting over fields. Over more general rings, however, this is no longer the case.

In this paper, we again carry out a reduction step to the rank 3 case, as in [DM07], this time for local Bézout domains R . We will establish our results for the group

$$\overline{U}_R^+ = \varphi_R(U_R^+) \leq \mathfrak{G}_A(\mathbb{K}),$$

where $\mathbb{K}$ is the field of fractions of R and φ_R is the group homomorphism

$$\varphi_R: \mathfrak{G}_A(R) \rightarrow \mathfrak{G}_A(\mathbb{K}),$$

naturally induced by the inclusion $R \hookrightarrow \mathbb{K}$. It was recently proven in [MM25] that, in this context, φ_R is actually injective, and hence $\overline{U}_R^+ \cong U_R^+$. However, we will not need this fact, and in fact our methods provide an alternative proof of the main result of *loc. cit.* in the 3-spherical simply laced case (see Corollary D below).

For each $\alpha \in \Phi$, we again write U_α for its image in $\mathfrak{G}_A(\mathbb{K})$, and we write $\overline{U}_{i,j}$ for $\varphi_R(\langle U_{\alpha_i}, U_{\alpha_j} \rangle)$. The reduction step to the rank 3 case which we obtain is then as follows.

Theorem B. *Let $(A_{ij})_{i,j \in I}$ be a 3-spherical generalised Cartan matrix. Let R be a local Bézout domain satisfying (co). Suppose that for every $J \subseteq I$ with $|J| = 3$, the subgroup*

$$U_{J,R}^+ := \mathfrak{G}_{A_J}(R) \cap U_R^+$$

of U_R^+ is the amalgamated product of its subgroups $U_{j_1,j_2} := \langle U_{\alpha_{j_1}}, U_{\alpha_{j_2}} \rangle$ along the canonical inclusions $U_{\alpha_{j_1}} \hookrightarrow U_{j_1,j_2}$, for all $j_1, j_2 \in J$. Then $\overline{U}_R^+$ is the amalgamated product of its subgroups $\overline{U}_{i_1,i_2}$ along the inclusions $U_{\alpha_{i_1}} \hookrightarrow \overline{U}_{i_1,i_2}$, for all $i_1, i_2 \in I$.

The condition (co) is the following mild assumption on R .

- (co) R has no quotient $\mathbb{F}_2$ if $A_{ij}A_{ji} = 2$ for some $i, j \in I$, and R has no quotient $\mathbb{F}_2$ or $\mathbb{F}_3$ if $A_{ij}A_{ji} = 3$ for some $i, j \in I$.

In combination with Theorem A, we then obtain a presentation for the subgroups $\overline{U}_R^+$ of 3-spherical simply laced Kac–Moody groups.

Corollary C. *Let R be a local Bézout domain such that its residue field is not $\mathbb{F}_2$. Let $(A_{ij})_{i,j \in I}$ be a simply laced generalised Cartan matrix. Then $\overline{U}_R^+ \leq \mathfrak{G}_A^{\min}(R)$ is the amalgamated product of its subgroups $\overline{U}_{i_1,i_2}$ along the canonical inclusions $U_{\alpha_{i_1}} \hookrightarrow \overline{U}_{i_1,i_2}$, for all $i_1, i_2 \in I$, if and only if A is 3-spherical.*

As alluded to above, Corollary C implies that under the same hypotheses, the canonical map $\varphi_R: \mathfrak{G}_A(R) \rightarrow \mathfrak{G}_A(\mathbb{K})$ is injective, thus providing an alternative proof in the 3-spherical simply laced case of the recent result [MM25, Theorem A] that φ_R is injective.

Corollary D. *Let A be a 3-spherical simply laced GCM and let R be a local Bézout domain such that its residue field is different from $\mathbb{F}_2$. Then the canonical map*

$$\varphi_R: \mathfrak{G}_A(R) \rightarrow \mathfrak{G}_A(\mathbb{K})$$

is injective.

The first part of the paper is dedicated to the proof of Theorem B. We now briefly outline some of the methods we will use to prove this result.

To obtain their presentation of $U_{\mathbb{K}}^+$ for A non-spherical and $\mathbb{K}$ a field, the authors of [DM07] make use [Tit86, Corollaire 1] (sometimes referred to as the *Tits lemma*) which, for groups G acting transitively on a chamber system C , characterises the existence of a presentation of G as the amalgamated product of the stabilisers of rank 2 residues in C , in terms of the simple connectedness of C .

The authors of [DM07] moreover introduce *filtrations* of chamber systems C , meaning a sequence $C_0 \subseteq C_1 \subseteq C_2 \subseteq \dots$ of subcomplexes of C such that $\bigcup_{n \in \mathbb{N}} C_n = C$, satisfying some additional properties, and show that under certain conditions, the simple connectedness of C is equivalent to the simple connectedness of C_0 (see Theorem 3.14 in *loc. cit.* or Proposition 4.4 below).

Since Kac–Moody groups over fields $\mathbb{K}$ have an associated building $\mathcal{C}^-(\mathfrak{G}_A(\mathbb{K})) = \mathfrak{G}_A(\mathbb{K})/B_{\mathbb{K}}^-$, which is a simply connected chamber system, the strategy of *loc. cit.* is to construct a suitable filtration of $\mathcal{C}^-(\mathfrak{G}_A(\mathbb{K}))$ such that $U_{\mathbb{K}}^+$ acts transitively on C_0 . A condition on $(C_n)_{n \in \mathbb{N}}$ required to apply their result and conclude that C_0 is simply connected, amounts to the rank 3 residues of C_0 being simply connected. In turn, by the Tits lemma, this condition is equivalent to the subgroups $U_{J\mathbb{K}}^+$ of $U_{\mathbb{K}}^+$ with $|J| = 3$ admitting a presentation as an amalgamated product of their rank 2 subgroups.

For any domain R with field of fractions $\mathbb{K}$, the Kac–Moody group $\mathfrak{G}_A^{\min}(R) := \varphi_R(\mathfrak{G}_A(R)) \leq \mathfrak{G}_A(\mathbb{K})$ also has an associated chamber system $\mathcal{C}^-(\mathfrak{G}_A^{\min}(R))$ on which it acts transitively. When R is a Bézout domain, it is moreover isomorphic to the building $\mathcal{C}^-(\mathfrak{G}_A(\mathbb{K}))$, by [MM25, Lemma 8.7]. Hence in this case, $\mathcal{C}^-(\mathfrak{G}_A^{\min}(R))$ is simply connected.

In the first part of this paper, we obtain a Birkhoff decomposition for $\mathfrak{G}_A^{\min}(R)$ for R a local ring. We then use this result to construct, under the assumptions of Theorem B, a filtration $(C_n)_{n \in \mathbb{N}}$ of $\mathcal{C}^-(\mathfrak{G}_A^{\min}(R))$, such that $\overline{U}_R^+$ acts transitively on C_0 , and which reduces the simple connectedness of C_0 to that of its rank 3 residues.

This article is structured as follows. In section 2, we introduce the necessary preliminaries on Kac–Moody groups which we will use throughout the rest of the paper. In section 3 we construct a Birkhoff decomposition of $\mathfrak{G}_A^{\min}(R)$, for R a local ring. Section 4 is dedicated to the proof of Theorem B: we first give the necessary background from [DM07], and then obtain the filtration required to carry out the reduction step to the rank 3 case, showing Theorem B. In section 5, we prove Theorem A, and in section 6 we conclude with the proofs of Corollary C and Corollary D.

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Declaration on the usage of AI. This work, from the initial research phase until the present written article, was completed without the use of artificial intelligence at any stage.

Notation. Throughout this article, for a group G and elements $x, y \in G$, we denote by $[x, y]$ the commutator $[x, y] := x^{-1}y^{-1}xy$. Moreover, we define $x^y := y^{-1}xy$.

2. PRELIMINARIES

2.1. Coxeter groups. For a reference on the notions introduced in this section, we refer to [AB08, Chapter 2]. A *Coxeter system* consists of a pair (W, S) , where S is a (finite) set of involutions which generate the *Coxeter group* W , the latter of which is given by a presentation

$$W := \langle s_i \in S \mid (s_i s_j)^{2m_{ij}} = 1 \rangle,$$

where $m_{ij} \in \mathbb{N} \cup \{\infty\}$ and $m_{ii} = 1$ for all i . Let $n = |S|$. We call $(m_{ij})_{1 \leq i, j \leq n}$ the *Coxeter matrix* associated to (W, S) . For any $w \in W$, there is a uniquely defined *length* $\ell(w)$ defined as d if $w = s_{i_1} \dots s_{i_d}$ and d is minimal for this property. We say that the word $\mathbf{w} = (s_{i_1}, \dots, s_{i_d})$ on the alphabet S is *reduced* if for the corresponding element $w := s_{i_1} \dots s_{i_d} \in W$, we have $\ell(w) = d$.

By Tits' solution to the word problem for Coxeter groups (see for example [AB08, Theorem 2.33]), any word $(s_{j_1}, \dots, s_{j_m})$ may be rewritten in reduced form by carrying out a sequence of the following two operations:

- (M1) Deleting a subword of the form (s_i, s_i) .
- (M2) Replacing a subword $\mathbf{t}(i, j) := (s_i, s_j, s_i, \dots)$ of length $m_{ij} < \infty$ by the word $\mathbf{t}(j, i) := (s_j, s_i, s_j, \dots)$ of length $m_{ji} = m_{ij}$.

We will denote by $\tau(i, j)$ the element $s_i s_j s_i \dots \in W$ corresponding to the word $\mathbf{t}(i, j)$.

A Coxeter system (W, S) has a naturally associated simplicial complex, called a *Coxeter complex*, on which W acts. It comes equipped with a set Φ of *roots* or *half spaces*, consisting of *simple roots* α_{s_i} for $s_i \in S$, and their images $w(\alpha_{s_i})$ for all $w \in W$. The root system Φ decomposes into a union $\Phi = \Phi^+ \sqcup \Phi^-$ of subsets Φ^+ and Φ^- consisting of respectively *positive* and *negative* roots. The sign of roots is characterised as follows.

Lemma 2.1 ([Hum90, Theorem 5.4]). *Let (W, S) be a Coxeter system with root system $\Phi = \Phi^+ \sqcup \Phi^-$. Let $\alpha_s \in \Phi^+$ for $s \in S$ denote the simple root associated to s . Then for any $w \in W$, $\ell(ws) < \ell(w)$ if and only if $w(\alpha_s) \in \Phi^-$.*

Let $J \subseteq S$. We say that $w \in W$ is *right J -reduced* if $\ell(ws_j) > \ell(w)$ for all $s_j \in J$. We denote moreover by W_J the subgroup of W generated by all $s_j \in J$. For any $w \in W$, the coset wW_J has a unique shortest element w_J . This is the unique element $w_J \in wW_J$ such that $\ell(w_J s_j) = \ell(w_J) + 1$ for all $s_j \in J$. The element w_J is right J -reduced. Moreover, for any $w' \in wW_J$, $\ell(w') = \ell(w_J) + \ell(w_J^{-1} w')$ (see [AB08, Proposition 2.20]).

2.2. Generalised Cartan matrices and associated Weyl group. Let $A = (A_{ij})_{i, j \in I}$ be a matrix with integer coefficients. It is called a *generalised Cartan matrix* (GCM) if for all $i \neq j \in I$, $A_{ii} = 2$, $A_{ij} \leq 0$ and $A_{ij} = 0 \iff A_{ji} = 0$.

A GCM is called *spherical* (equivalently, of *finite type*) if it is a Cartan matrix. We say that A is *n-spherical* if $A_J := (A_{ij})_{i, j \in J}$ is spherical, for all $J \subseteq I$ with $|J| \leq n$. The size $|I|$ of I is called the *rank* of A . In this paper, all GCM's A are assumed to be 2-spherical.

A GCM is called *decomposable* if it can be written as a block-diagonal matrix $\begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$ after applying a permutation of its index set I .

A GCM is called *simply laced* if for all $J \subseteq I$ of rank 2, $A_J = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ or $A_J = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. Simply laced GCM's are 2-spherical. The Cartan matrix of type A_3 is given by

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Let $Q := \bigoplus_{i \in I} \mathbb{Z}\alpha_i$ be the free abelian group generated by symbols α_i , for all $i \in I$. They are called the *simple roots* associated to A . We denote $\Pi := \{\alpha_i \mid i \in I\}$. Let $\mathcal{W} \leq \text{GL}(Q)$ be the group generated by the *simple reflections*

$$s_i: Q \rightarrow Q, \quad \alpha_j \mapsto \alpha_j - A_{ij}\alpha_i.$$

It is called the *Weyl group* of A .

The pair $(\mathcal{W}, S)$, where $S := \{s_i \mid i \in I\}$, is a Coxeter system, with associated Coxeter matrix $(m_{ij})_{i,j \in I}$ where for $i \neq j$, we have $m_{ij} = 2, 3, 4, 6$, or ∞ , respectively if and only if $A_{ij}A_{ji} = 0, 1, 2, 3$ or ≥ 4 . Note that the GCM A is spherical if and only if $\mathcal{W}$ is finite, and A is 2-spherical if and only if $m_{ij} < \infty$ for all $i, j \in I$. The set $\Phi := \mathcal{W} \cdot \Pi$ is called the set of *real roots* associated to A , and it is in one-to-one correspondence with the set of roots of the Coxeter complex of $(\mathcal{W}, S)$.

For any $J \subseteq I$, we define $\Phi^\pm(J) := \Phi^\pm \cap \bigoplus_{j \in J} \mathbb{Z} \alpha_j$.

A pair of distinct roots $\alpha, \beta \in \Phi$ is called *prenilpotent* if there are some $w_1, w_2 \in \mathcal{W}$ such that $\{w_1\alpha, w_2\beta\} \subseteq \Phi^+$ and $\{w_2\alpha, w_1\beta\} \subseteq \Phi^-$. Then the set $] \alpha, \beta[_{\mathbb{N}} := (\mathbb{N} \alpha_i + \mathbb{N} \alpha_j) \cap \Phi$ is finite (see [Mar18, Definition 7.41]).

2.3. The Kac–Moody groups $\mathfrak{G}_A(R)$ and $\mathfrak{G}_A^{\min}(R)$. In this section, we briefly introduce the Kac–Moody groups $\mathfrak{G}_A(R)$ and $\mathfrak{G}_A^{\min}(R)$ under consideration in this article. We refer respectively to [Mar18, Chapter 7] and to [Mar18, Section 8.8] for the precise definitions of $\mathfrak{G}_A(R)$ and $\mathfrak{G}_A^{\min}(R)$. Throughout this section, let R be a commutative ring with unit.

The groups $\mathfrak{G}_A(R)$. Let $A = (A_{ij})_{i,j \in I}$ be a GCM. The (“*simply connected*”) *minimal Kac–Moody group $\mathfrak{G}_A(R)$ of type A over R* is given by a presentation, with generators depending on the real roots $\Phi = \mathcal{W} \cdot \{\alpha_i \mid i \in I\}$ associated to A . We refer to [Mar18, Definition 7.47 & Example 7.11] for the precise definition (see also [Tit87]). For our purposes, the relevant information is the following.

For every $\alpha \in \Phi$, we specify a copy $\mathfrak{U}_\alpha(R) \cong (R, +)$ of the additive group of R , and denote the isomorphism by

$$x_\alpha: (R, +) \rightarrow \mathfrak{U}_\alpha(R).$$

We denote by $\mathfrak{St}_A(R)$ the group defined as the free product of all the $\mathfrak{U}_\alpha(R)$ as α runs through Φ , quotiented out by the relations

$$(2.1) \quad [x_\alpha(r), x_\beta(s)] = \prod_{\gamma} x_\gamma(C_{ij}^{\alpha\beta} r^i s^j), \quad \text{for all } \{\alpha, \beta\} \subseteq \Phi \text{ prenilpotent and all } r, s \in R,$$

where $\gamma = i\alpha + j\beta$ enumerates $] \alpha, \beta[_{\mathbb{N}}$ in a fixed order, and the constants $C_{ij}^{\alpha\beta}$ are integers as given by [Mar18, Proposition 7.43]. The group $\mathfrak{G}_A(R)$ is then defined in terms of a presentation, generated by the group $\mathfrak{St}_A(R)$, subject to some additional relations (see [Mar18, Definition 7.47]).

Throughout, we set $G_R := \mathfrak{G}_A(R)$, and distinguish the following subgroups of G_R . We identify

$$U_\alpha = U_{\alpha R} := \mathfrak{U}_\alpha(R), \text{ for any } \alpha \in \Phi,$$

with their images in G_R , and define moreover

$$U_{AR}^\pm := U_R^\pm := \langle U_{\alpha R} \mid \alpha \in \Phi^\pm \rangle.$$

If $A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$ is decomposable, then $U_{AR}^\pm \cong U_{A_1 R}^\pm \times U_{A_2 R}^\pm$.

Inside G_R , there is a *torus* $T_R \cong (R^\times)^{|I|}$, and it follows from the defining relations of G_R that T_R normalises $U_R^\pm$. We define the *Borel subgroups* $B_R^\pm \leq G_R$ as

$$B_R^\pm := U_R^\pm T_R.$$

For any ring R , we will write $B_2^+(R)$ (resp. $B_2^-(R)$) for the subgroup of upper (resp. lower) triangular matrices of $\mathrm{SL}_2(R)$, and $U_2^\pm(R)$ for the subgroup of $B_2^\pm(R)$ consisting of unipotent matrices.

For $i \in I$, we define moreover

$$G_{iR} := \langle U_{\alpha_i R}, U_{-\alpha_i R} \rangle \leq G_R \quad \text{and} \quad P_{iR}^\pm := \langle G_{iR}, B_R^\pm \rangle \leq G_R,$$

and for $J \subseteq I$,

$$U_{JR}^\pm := \langle U_{\alpha R} \mid \alpha \in \Phi^\pm(J) \rangle, \quad \text{and} \quad G_{JR} := \langle G_{jR} \mid j \in J \rangle, \quad \text{and} \quad P_{JR}^\pm := \langle G_{JR}, B_R^\pm \rangle.$$

Moreover, for any A , the torus T_R normalises each G_{iR} . The following fact will be useful to us.

Lemma 2.2 ([MM25, Lemma 4.3]). *Let A be 2-spherical and suppose R satisfies (co) in case A is non-spherical. Then for any $i \in I$ and $\varepsilon \in \{\pm\}$,*

$$G_{iR}U_R^\varepsilon = U_R^\varepsilon G_{iR} \quad \text{and} \quad G_{iR}B_R^\varepsilon = B_R^\varepsilon G_{iR}.$$

The result below will be essential for our construction of filtrations fitting the requirements of the work in [DM07].

Lemma 2.3 ([MM25, Proposition 4.2]). *Let A be a 2-spherical GCM and suppose that R satisfies (co). Then $U_R^\pm = \langle U_{\pm\alpha_i R} \mid i \in I \rangle$. In particular, G_R is generated by its subgroups G_{iR} and T_R , for $i \in I$.*

For any $i \in I$, we define $\tilde{s}_i := x_{\alpha_i}(1)x_{-\alpha_i}(1)x_{\alpha_i}(1)$. One can prove (see [Mar18, Proposition 7.57]) that for any $w \in \mathcal{W}$ with reduced expression $(s_{i_1}, \dots, s_{i_n})$, the element $\tilde{w} := \tilde{s}_{i_1} \dots \tilde{s}_{i_n} \in G_R$ depends only on w and not on the chosen expression $(s_{i_1}, \dots, s_{i_n})$. In particular, the subgroup $N_R := \langle \tilde{s}_i, T_R \mid i \in I \rangle \leq G_R$ surjects onto $\mathcal{W}$, with kernel T_R . One can also show that when $\mathbb{K}$ is a field, $P_{J\mathbb{K}}^\pm = B_{\mathbb{K}}^\pm \mathcal{W}_J B_{\mathbb{K}}^\pm$, see [Mar18, §B.3]. Moreover, for any $w \in \mathcal{W}$ and any $\alpha \in \Phi$,

$$\tilde{w}U_\alpha \tilde{w}^{-1} = U_{w\alpha}.$$

Lemma 2.4. *Let A be a GCM with Weyl group $\mathcal{W}$ and with real roots $\Phi = \Phi^+ \sqcup \Phi^-$. Let $J \subseteq I$ and let $w \in \mathcal{W}$ be right J -reduced. Then*

$$\tilde{w}U_{JR}^+ \tilde{w}^{-1} \subseteq B_R^+.$$

Proof. The conjugate $\tilde{w}U_{JR}^+ \tilde{w}^{-1}$ is generated by $x_{w\alpha_j}(\lambda)$ for $j \in J$ and $\lambda \in R$. Then by Lemma 2.1, $w\alpha_j \in \Phi^+$ for any $j \in J$, since w is right J -reduced. It follows that $\tilde{w}U_{JR}^+ \tilde{w}^{-1} \subseteq B_R^+$. $\square$

An important feature of the theory of Kac–Moody groups over fields is the fact that they have an associated (twin) BN -pair, and hence act on an associated (twin) building (see also section 4.1). In particular, for $\mathbb{K}$ a field and $\varepsilon \in \{\pm\}$, $G_{\mathbb{K}}$ has *Bruhat decompositions*

$$(2.2) \quad G_{\mathbb{K}} = \coprod_{w \in \mathcal{W}} B_{\mathbb{K}}^\varepsilon \tilde{w} B_{\mathbb{K}}^\varepsilon,$$

and *Birkhoff decompositions*

$$(2.3) \quad G_{\mathbb{K}} = \coprod_{w \in \mathcal{W}} B_{\mathbb{K}}^{-\varepsilon} \tilde{w} B_{\mathbb{K}}^\varepsilon = \coprod_{w \in \mathcal{W}} U_{\mathbb{K}}^{-\varepsilon} \tilde{w} B_{\mathbb{K}}^\varepsilon,$$

see [Mar18, Corollary 7.70 & §B.3].

In particular, from the Bruhat decompositions it follows that

$$\mathrm{SL}_2(\mathbb{K}) = B_2^\varepsilon(\mathbb{K}) \sqcup B_2^\varepsilon(\mathbb{K}) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} B_2^\varepsilon(\mathbb{K}),$$

for any field $\mathbb{K}$ and $\varepsilon \in \{\pm\}$. The double cosets in the Bruhat and Birkhoff decompositions satisfy the following properties (see [AB08, Proposition 6.36 & Proposition 6.77]). Let $B^\varepsilon := B_{\mathbb{K}}^\varepsilon$ and $U^\varepsilon := U_{\mathbb{K}}^\varepsilon$, and write moreover w for $\tilde{w} \in G_{\mathbb{K}}$.

$$(2.4) \quad \begin{aligned} B^\varepsilon w B^\varepsilon s B^\varepsilon &\subseteq B^\varepsilon w B^\varepsilon \cup B^\varepsilon w s B^\varepsilon, \quad \text{and} \\ B^\varepsilon w B^\varepsilon s B^\varepsilon &= B^\varepsilon w s B^\varepsilon \iff \ell(ws) = \ell(w) + 1, \end{aligned}$$

and

$$(2.5) \quad \begin{aligned} U^{-\varepsilon} w B^{\varepsilon} s B^{\varepsilon} &\subseteq U^{-\varepsilon} w B^{\varepsilon} \cup U^{-\varepsilon} w s B^{\varepsilon}, \quad \text{and} \\ U^{-\varepsilon} w B^{\varepsilon} s B^{\varepsilon} &= U^{-\varepsilon} w s B^{\varepsilon} \iff \ell(ws) = \ell(w) - 1, \end{aligned}$$

Finally, G_R possesses a *Cartan–Chevalley involution* $\omega_R \in \text{Aut}(G_R)$ which preserves T_R and all $\tilde{s}_i$, and which sends $U_{\alpha R}$ to $U_{-\alpha R}$ for all $\alpha \in \Phi$.

The groups $\mathfrak{G}_D^{\min}(R)$. Throughout this section, we assume that R is a local ring. For $A = (A_{ij})_{i,j \in I}$ a GCM, there is also a more involved notion of the (*minimal*) *Tits functor* $\mathfrak{G}_A^{\min}$, as defined in [Mar18, Definition 8.126]. It has the property that for any $i \in I$, there is a group functor morphism $\varphi_i: \text{SL}_2 \rightarrow \mathfrak{G}_A^{\min}$, and moreover there is a group functor morphism

$$\varphi: \mathfrak{G}_A \rightarrow \mathfrak{G}_A^{\min}.$$

We write $\varphi_{iR}: \text{SL}_2(R) \rightarrow \mathfrak{G}_A^{\min}(R)$ and $\varphi_R: \mathfrak{G}_A(R) \rightarrow \mathfrak{G}_A^{\min}(R)$ for the respective group homomorphisms naturally induced by φ_i and φ . Then for any $r \in R$, and any $i \in I$, the map φ_R has the property that

$$\varphi_{iR}\left(\begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}\right) = \varphi_R(x_{\alpha_i}(r)) \quad \text{and} \quad \varphi_{iR}\left(\begin{pmatrix} 1 & 0 \\ -r & 1 \end{pmatrix}\right) = \varphi_R(x_{-\alpha_i}(r)).$$

In particular,

$$\overline{G}_{iR} \cong \langle \left(\begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -r & 1 \end{pmatrix} \mid r \in (R, +) \right) \rangle \cong \text{SL}_2(R),$$

where the latter isomorphism holds because R is a local ring.

Additionally, for any injective ring morphism $R_1 \rightarrow R_2$, the induced group homomorphism $\mathfrak{G}_A^{\min}(R_1) \rightarrow \mathfrak{G}_A^{\min}(R_2)$ is injective.

Since R is a local ring, $\mathfrak{G}_A^{\min}(R)$ is the surjective image $\varphi_R(\mathfrak{G}_A(R))$ of $\mathfrak{G}_A(R)$, and the work in [MM25] is dedicated to investigating when φ_R is injective, i.e. to when the groups $\mathfrak{G}_A(R)$ and $\mathfrak{G}_A^{\min}(R)$ coincide. This was already known by Tits' work in [Tit87] to be the case when $R = \mathbb{K}$ is a field, and is moreover proven to be the case when R satisfies (co) and is a *valuation ring*, i.e. a local ring which is moreover a Bézout domain in [MM25, Theorem A].

When R is a domain with field of fractions $\mathbb{K}$, there is a natural map

$$\psi: \mathfrak{G}_A(R) \rightarrow \mathfrak{G}_A(\mathbb{K}),$$

induced by the canonical inclusions $\mathfrak{U}_{\alpha}(R) \hookrightarrow \mathfrak{U}_{\alpha}(\mathbb{K})$, for all $\alpha \in \Phi$.

Since $\mathfrak{G}_A(\mathbb{K}) \cong \mathfrak{G}_A^{\min}(\mathbb{K})$, the group $\mathfrak{G}_A^{\min}(R)$ injects into $\mathfrak{G}_A(\mathbb{K})$, and under this identification, the map

$$\varphi_R: \mathfrak{G}_A(R) \rightarrow \mathfrak{G}_A^{\min}(R) \subseteq \mathfrak{G}_A(\mathbb{K})$$

coincides with ψ , so that $\mathfrak{G}_A^{\min}(R)$ is just the image of $\mathfrak{G}_A(R)$ in $\mathfrak{G}_A(\mathbb{K})$ induced by the inclusion $R \hookrightarrow \mathbb{K}$.

Throughout the rest of this article, we will write $G_R^{\min} := \mathfrak{G}_A^{\min}(R)$. Moreover, for a subgroup $H \leq G_R$, we will denote by $\overline{H}$ its image under φ_R in $G_R^{\min}$, e.g. $\overline{U}_R^{\pm} := \varphi_R(U_R^{\pm})$.

Note that φ_R preserves T_R as well as the elements $\tilde{s}_i$ for all $i \in I$, and the restriction of φ_R to any root subgroup U_{α} is injective. In particular, we also write T_R , $\tilde{s}_i$ and U_{α} for their respective images under φ_R .

2.4. Some facts about G_R and $G_R^{\min}$. Unless otherwise stated, in this article we let R be a local ring and denote by $R \rightarrow k$ the projection onto its residue field $k := R/L$, where L is the maximal ideal of R . We then write

$$(2.6) \quad p: G_R^{\min} \rightarrow G_k^{\min} \cong G_k$$

for the group homomorphism induced by functoriality.

If we denote respectively by $\overline{U}_L^{\pm}$ and $\overline{B}_L^{\pm}$ the kernels of

$$p|_{\overline{U}_R^{\pm}}: \overline{U}_R^{\pm} \rightarrow U_k^{\pm} \quad \text{and} \quad p|_{\overline{B}_R^{\pm}}: \overline{B}_R^{\pm} \rightarrow B_k^{\pm},$$

one has the following.

Lemma 2.5. *Let R and $p: G_R^{\min} \rightarrow G_k$ be as above. Then*

$$(2.7) \quad \text{Ker}(p) = \overline{U}_L^- \overline{B}_L^+ = \overline{U}_L^+ \overline{B}_L^-.$$

Proof. By [MM25, Proposition 5.3] and surjectivity of φ_R , one has that $\text{Ker}(p) \subseteq \overline{U}_L^- \overline{B}_L^+$. If $g = u_- b_+ \in \text{Ker}(p)$ with $u_- \in \overline{U}_L^-$ and $b_+ \in \overline{B}_L^+$, then $u_-^{-1}g \in \text{Ker}(p)$ and hence $b_+ \in \overline{B}_L^+$. The result now follows from the fact that $\text{Ker}(p)^{-1} = \text{Ker}(p)$ and that T_R normalises $\overline{U}_L^+$. $\square$

The following lemma is particularly useful.

Lemma 2.6 ([MM25, Lemma 5.10]). *Let $A = (A_{ij})_{i,j \in I}$ be 2-spherical, and let R be a local ring satisfying (co) in case A is not spherical. Let π_R denote the composition of $\varphi_R: G_R \rightarrow G_R^{\min}$ and $p: G_R^{\min} \rightarrow G_k$. Then, for $J \subseteq I$,*

$$G_{JR} \cap \pi_R^{-1}(B_k^- B_k^+) \subseteq T_R U_{JR}^- U_{JR}^+.$$

We also record the following lemma for convenience. Here $U_2^\pm(L)$ denotes the kernel of the canonical map $\text{SL}_2(R) \rightarrow \text{SL}_2(k)$ induced by $R \rightarrow k$, restricted to $U_2^\pm(R)$.

Lemma 2.7 ([MM25, Lemma 5.2]). *Let $i \in I$ and $g \in \overline{G}_{iR}$. If $p(g) \in \varphi_{ik}(B_2^+(k))$, then $g \in \varphi_{iR}(U_2^-(L)B_2^+(R))$. Similarly, if $p(g) \in \varphi_{ik}(B_2^-(k))$, then $g \in \varphi_{iR}(U_2^+(L)B_2^-(R))$.*

When R is a domain, under the identification $G_{\mathbb{K}}^{\min} \cong G_{\mathbb{K}}$, we have

$$(2.8) \quad \overline{U}_R^\pm \cap \overline{B}_R^\mp \subseteq U_{\mathbb{K}}^\pm \cap B_{\mathbb{K}}^\mp = \{1\},$$

where the last equality follows from the Birkhoff decomposition of $G_{\mathbb{K}}$ (see [MM25, eq. (2.1)]). This condition also implies that $T_R \cap \overline{U}_R^\pm = \{1\}$.

Lemma 2.8. *Let $A = (A_{ij})_{i,j \in I}$ be a 2-spherical GCM. Let R be a local domain satisfying (co). Then for any $J \subseteq I$,*

$$(2.9) \quad \overline{U}_R^+ \cap \overline{P}_{JR}^- = \overline{U}_{JR}^+,$$

Proof. The inclusion $\overline{U}_R^+ \subseteq \overline{U}_R^+ \cap \overline{P}_{JR}^-$ is trivial. For the opposite inclusion, note first that

$$\overline{U}_R^+ \cap \overline{P}_{JR}^- = \overline{U}_R^+ \cap \overline{P}_{JR}^+ \cap \overline{P}_{JR}^- = \overline{U}_R^+ \cap T_R \overline{G}_{JR},$$

where the latter equality follows from [MM25, Proposition 5.12]. Suppose that $g \in \overline{U}_R^+$ satisfies $\varphi_R(g) \in T_R \overline{G}_{JR}$. Let $t \in T_R$ and $g_J \in G_{JR}$ be such that $\varphi_R(g) = \varphi_R(tg_J)$. Then

$$p(\varphi_R(tg_J)) = p(\varphi_R(g)) \in U_k^+ \subseteq B_k^- B_k^+.$$

In particular, by Lemma 2.6 it follows that $tg_J \in T_R U_{JR}^- U_{JR}^+$. Choose $t' \in T_R$ and $u_\pm \in U_{JR}^\pm$ such that $tg_J = t'u_- u_+$. In particular, $\varphi_R(g) = \varphi_R(t'u_- u_+)$. Then, by (2.8),

$$\varphi_R(t'^{-1}gu_+^{-1}) = \varphi_R(u_-) \in T_R \overline{U}_R^+ \cap \overline{U}_{JR}^- = \{1\}.$$

Hence $\varphi_R(t') = \varphi_R(gu_+^{-1}) \in T_R \cap \overline{U}_R^+ = \{1\}$, and we obtain that $\varphi_R(g) = \varphi_R(u_+) \in \overline{U}_{JR}^+$. $\square$

3. BIRKHOFF DECOMPOSITION OF $G_R^{\min}$

Throughout this section, we fix a 2-spherical GCM $A = (A_{ij})_{i,j \in I}$, and R a local ring. We aim to construct a decomposition of $G_R^{\min}$ which generalises the Birkhoff decompositions in the field case.

Definition 3.1. Let $g \in G_R^{\min}$. A *Birkhoff expression* of g is a decomposition of the form

$$g = u_+ g_1 \dots g_d b_-,$$

with $u_+ \in \overline{U}_R^+$ and $b_- \in \overline{B}_R^-$, and such that for all $1 \leq j \leq d$, there is some $i_j \in I$ such that $g_j \in \overline{G}_{i_j R}$. The *type* of such a Birkhoff expression of g is the tuple $(s_{i_1}, \dots, s_{i_d})$, for $s_i \in \mathcal{W}$. The Birkhoff expression $g = u_+ g_1 \dots g_d b_+$ is *reduced* if d is minimal among the lengths of types of Birkhoff expressions of g .

For A a GCM, let $(m_{ij})_{i,j \in I}$ denote its Coxeter matrix and for any $i, j \in I$, let $\mathbf{t}(i, j)$ and $\tau(i, j)$ be as in section 2.1. The following result is a construction of subsets of $G_R^{\min}$ generalising the double cosets appearing in the Birkhoff decomposition of a Kac-Moody group over a field.

Proposition 3.2. Let R be a local ring satisfying (co) in case A is not spherical. Let $p: G_R^{\min} \rightarrow G_k$ be as in (2.6) and let $\mathbf{w} = (s_{i_1}, \dots, s_{i_d})$ be a word in the generators $\{s_i \mid i \in I\}$ of $\mathcal{W}$. Define

$$B(\mathbf{w}) := \left\{ g \in G_R^{\min} \mid g \text{ has a reduced Birkhoff expression of type } (s_{i_1}, \dots, s_{i_d}) \right\}.$$

Then the following hold:

- (i) $g \in B(\mathbf{w})$ implies that $p(g) \in U_k^+ \tilde{w} B_k^-$,
- (ii) Let $\mathbf{w}' = (s_{j_1}, \dots, s_{j_d})$ differ from $\mathbf{w}$ by a finite sequence of operation of type (M2). Then $B(\mathbf{w}) = B(\mathbf{w}')$.
- (iii) The set $B(\mathbf{w})$ is non-empty if and only if $\mathbf{w}$ is reduced.

Proof. (i). Let $\mathbf{w} = (s_{i_1}, \dots, s_{i_d})$. We prove the statement by induction on d . Let $g \in B(\mathbf{w})$. Suppose $d = 0$. Then $g \in \overline{U}_R^+ \overline{B}_R^-$. It then follows by definition that $p(g) \in U_k^+ B_k^-$.

Suppose now that for any $\mathbf{w} = (s_{i_1}, \dots, s_{i_d})$ with $d \leq n$ and any $g \in B(\mathbf{w})$, one has $p(g) \in U_k^+ \tilde{w} B_k^-$, where $w = s_{i_1} \dots s_{i_d} \in \mathcal{W}$.

Let then $\mathbf{w} = (s_{i_1}, \dots, s_{i_{n+1}})$ and $g \in B(\mathbf{w})$, i.e.

$$g = u_+ g_1 \dots g_{n+1} b_-, \text{ for some } u_+ \in \overline{B}_R^-, \quad g_j \in \overline{G}_{i_j R}, \quad \text{and } b_- \in \overline{B}_R^-,$$

with $n+1$ minimal amongst the lengths of Birkhoff expressions of g . By Lemma 2.2, $\overline{B}_R^- \overline{G}_{i_R} = \overline{G}_{i_R} \overline{B}_R^-$ and $\overline{G}_{i_R} \overline{U}_R^+ = \overline{U}_R^+ \overline{G}_{i_R}$, for any $i \in I$. Hence for all $j \in \{1, \dots, n+1\}$,

$$g_j \notin \varphi_{i_j R}(U_2^+(R) B_2^-(R)) \subseteq \overline{U}_R^+ \overline{B}_R^-,$$

by minimality of $n+1$. Then Lemma 2.7 implies that $p(g_j) \notin \varphi_{i_j k}(B_2^-(k))$. It follows in particular that

$$p(g_{n+1}) \in \varphi_{i_{n+1} k} \left(B_2^-(k) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} B_2^-(k) \right) \subseteq B_k^- \tilde{s}_{i_{n+1}} B_k^-.$$

Additionally, writing $\mathbf{w}' := (s_{i_1}, \dots, s_{i_n})$, we have

$$g' := u_+ g_1 \dots g_n \in B(\mathbf{w}'),$$

since the above decomposition not being a reduced Birkhoff expression of g' , would contradict the reducedness of the Birkhoff expression $u_+ g_1 \dots g_{n+1} b_-$ of $g = u_+ g_1 \dots g_{n+1} b_- = g' g_{n+1} b_-$. By the induction hypothesis,

$$p(g') \in U_k^+ \tilde{s}_{i_1} \dots \tilde{s}_{i_n} B_k^-.$$

Combining the above then implies by (2.5) that

$$p(g) = p(g' g_{n+1} b_-) \in U_k^+ \tilde{s}_{i_1} \dots \tilde{s}_{i_n} B_k^- \tilde{s}_{i_{n+1}} B_k^- \subseteq U_k^+ \{ \tilde{s}_{i_1} \dots \tilde{s}_{i_n}, \tilde{s}_{i_1} \dots \tilde{s}_{i_{n+1}} \} B_k^-.$$

Suppose that $p(g) \in U_k^+ \tilde{s}_{i_1} \dots \tilde{s}_{i_n} B_k^-$. Then by Lemma 2.5,

$$g \in \overline{U}_R^+ \tilde{s}_{i_1} \dots \tilde{s}_{i_n} \text{Ker}(p) \overline{B}_R^- \subseteq \overline{U}_R^+ \tilde{s}_{i_1} \dots \tilde{s}_{i_n} \overline{U}_L^+ \overline{B}_R^- \subseteq \overline{U}_R^+ \overline{G}_{i_1 R} \dots \overline{G}_{i_n R} \overline{B}_R^-,$$

where the last inclusion follows from [MM25, Lemma 5.1], which implies that $\overline{G}_{i_R} \overline{U}_L^+ = \overline{U}_L^+ \overline{G}_{i_R}$ for any $i \in I$. But this is a contradiction with the minimality of $n+1$, and hence

$$p(g) \in U_k^+ \tilde{s}_{i_1} \dots \tilde{s}_{i_{n+1}} B_k^-,$$

as desired. This concludes the proof of (i).

(ii). Suppose $\mathbf{w}' = (s_{j_1}, \dots, s_{j_d})$ is obtained from $\mathbf{w} = (s_{i_1}, \dots, s_{i_d})$ by applying (M2) n times, for some $n \geq 1$. Let $w \in \mathcal{W}$ be the associated group element. Let $g \in B(\mathbf{w})$. By (i),

$$p(g) \in U_k^- \tilde{s}_{i_1} \dots \tilde{s}_{j_d} B_k^+ = U_k^- \tilde{w} B_k^+ = U_k^- \tilde{s}_{j_1} \dots \tilde{s}_{j_d} B_k^-.$$

In particular, since $\overline{U}_R^+ \tilde{w} \overline{B}_R^-$ surjects onto $U_k^+ \tilde{w} B_k^-$, we have as in the previous paragraph that

$$g \in \overline{U}_R^+ \tilde{s}_{j_1} \dots \tilde{s}_{j_d} \text{Ker}(p) \overline{B}_R^- \subseteq \overline{U}_R^+ \tilde{s}_{j_1} \dots \tilde{s}_{j_d} \overline{U}_L^+ \overline{B}_R^- \subseteq \overline{U}_R^+ \overline{G}_{j_1} \dots \overline{G}_{j_d} \overline{B}_R^-,$$

by Lemma 2.5 and [MM25, Lemma 5.1]. Hence g has a Birkhoff expression of type $(s_{j_1}, \dots, s_{j_d})$. By the fact that d is the minimal length for which a Birkhoff expression of g exists, it follows that this Birkhoff expression is reduced as well. Thus $g \in B(\mathbf{w}')$. The converse inclusion follows similarly. Hence we obtain (ii).

(iii). Suppose $\mathbf{w} = (s_{i_1}, \dots, s_{i_d})$ is not reduced. Let $g \in B(\mathbf{w})$. By Tits' solution to the word problem, there is an integer $n \geq 1$ such that, after applying some n operations of type (M2), the obtained word $\mathbf{w}' = (s_{j_1}, \dots, s_{j_d})$ has a subword of the form (s_i, s_i) . But then, from the fact that $B(\mathbf{w}) = B(\mathbf{w}')$ by (ii), it follows that

$$g \in \overline{U}_R^- \overline{G}_{j_1 R} \dots \overline{G}_{i R} \overline{G}_{i R} \dots \overline{G}_{j_d R} \overline{B}_R^- = \overline{U}_R^+ \overline{G}_{j_1 R} \dots \overline{G}_{i R} \dots \overline{G}_{j_d R},$$

where the equality holds because $\overline{G}_{i R} \overline{G}_{i R} = \overline{G}_{i R}$. But this would mean that g has an expression $g = u_+ g_1 \dots g_{d-1} b_-$, for some $g_\ell \in \overline{G}_{\ell R}$ and elements $u_+ \in \overline{U}_R^+$ and $b_- \in \overline{B}_R^-$. This is a contradiction with reducedness of the expression of g of type $\mathbf{w}$. Hence $B(\mathbf{w})$ is empty if $\mathbf{w}$ is not reduced.

Conversely, if $\mathbf{w} = (s_{i_1}, \dots, s_{i_d})$ is reduced, then $g := \tilde{s}_{i_1} \dots \tilde{s}_{i_n}$ is a reduced Birkhoff expression of g . Indeed, by definition, $p(\tilde{s}_{i_1} \dots \tilde{s}_{i_n}) \in U_k^- \tilde{w} B_k^+$ with $w = s_{i_1} \dots s_{i_n}$. Since $\mathbf{w}$ is reduced, it follows that $\ell(w) = n$. If g had a Birkhoff expression of shorter length, we would have by (i) that $p(g) \in U_k^+ w' B_k^-$ with $\ell(w') < \ell(w)$, a contradiction with the Birkhoff decomposition of G_k . Hence $g \in B(\mathbf{w})$. $\square$

It follows from Proposition 3.2 that for any $w \in \mathcal{W}$, the set $B(\mathbf{w})$ does not depend on the chosen reduced expression $\mathbf{w}$ of w . We then denote

$$(3.1) \quad B(w) := B(\mathbf{w}), \quad \text{for any reduced expression } \mathbf{w} = (s_{i_1}, \dots, s_{i_d}) \text{ of } w \in \mathcal{W}.$$

Theorem 3.3. *Let R be a local ring satisfying (co) in case A is not spherical. Then $G_R^{\min}$ has a decomposition*

$$(3.2) \quad G_R^{\min} = \coprod_{w \in \mathcal{W}} B(w).$$

Moreover,

$$(3.3) \quad B(w) = p^{-1}(U_k^+ \tilde{w} B_k^-) = \overline{U}_R^+ \tilde{w} \overline{B}_R^- \text{Ker}(p).$$

Proof. By Lemma 2.3, $G_R^{\min}$ is generated by the groups T_R and $\overline{G}_{i R}$ for $i \in I$. Since T_R normalises each $\overline{G}_{i R}$, for every element $g \in G_R^{\min}$ there is some sequence $(i_1, \dots, i_d) \in I^d$ such that

$$g \in \overline{G}_{i_1 R} \dots \overline{G}_{i_d R} T_R,$$

and hence g has a Birkhoff decomposition. In particular, $g \in B(w)$ for some $w \in \mathcal{W}$. It follows that $G_R^{\min} = \bigcup_{w \in \mathcal{W}} B(w)$.

Let $w \neq w' \in \mathcal{W}$ and suppose $g \in B(w) \cap B(w')$. Then by Proposition 3.2(i),

$$p(g) \in U_k^+ \tilde{w} B_k^- \cap U_k^+ \tilde{w}' B_k^-,$$

a contradiction by the Birkhoff decomposition of G_k . Hence $B(w) \cap B(w') = \emptyset$. The decomposition (3.2) follows. By Proposition 3.2(i), $B(w)$ surjects onto $U_k^- \tilde{w} B_k^+$, and hence $B(w) \subseteq p^{-1}(U_k^- \tilde{w} B_k^+)$. Conversely, if $g \notin B(w)$, then by (3.2), $g \in B(w')$ for some $w' \neq w$. Then again

by Proposition 3.2(i), $p(g) \in U_k^- \tilde{w}' B_k^+$, and hence $p(g) \notin U_k^- \tilde{w} B_k^+$, by the Birkhoff decomposition of G_k . In particular, $g \notin p^{-1}(U_k^- \tilde{w} B_k^+)$. This shows (3.3). $\square$

Remark 3.4. Note that using the Cartan–Chevalley involution, $G_R^{\min}$ also has a decomposition $G_R^{\min} = \coprod_{w \in \mathcal{W}} \tilde{B}(w)$, where, for $w = s_{i_1} \dots s_{i_d}$ of length d ,

$$\tilde{B}(w) := p^{-1}(U_k^- \tilde{w} B_k^+) = \overline{U}_R^- \tilde{w} \overline{B}_R^+ \text{Ker}(p).$$

4. RESIDUAL FILTRATIONS OF THE CHAMBER SYSTEM OF $G_R^{\min}$

In this section, we consider the chamber system associated to $G_R^{\min}$. In particular, we will construct a *filtration* of this chamber system, and use the results from [DM07] to reduce the question of obtaining a presentation of an arbitrary $\overline{U}_R^+$ of simply laced type to obtaining a presentation for $\overline{U}_R^+$ of type A_3 (in case R is a local Bézout domain).

4.1. Buildings and chamber systems. A *chamber system* is a set C together with a collection $(\sim_i)_{i \in I}$ of equivalence relations on C , such that $c \sim_i d$ and $c \sim_j d$ if and only if $i = j$ or $c = d$. Elements of C are called *chambers*, and two chambers $c, d \in C$ are called *i -adjacent* if $c \sim_i d$. A chamber system is said to be *connected* if any two chambers $c, d \in C$ can be joined by a *gallery*

$$c = c_1 \sim_{i_1} c_2 \sim_{i_2} \dots \sim_{i_k} c_{k+1} = d.$$

The *type* of such a gallery is the tuple $(i_1, \dots, i_k)$. Let $J \subseteq I$. A *J -gallery* is a gallery $c \sim_{j_1} \dots \sim_{j_k} d$ such that $j_1, \dots, j_k \in J$. For $c \in C$, the *J -residue* of c is the set $\mathcal{R}_J(c)$ of chambers $d \in C$ which can be joined to c by a J -gallery. Then $(\mathcal{R}_J(c), (\sim_j)_{j \in J})$ is also a chamber system.

Example 4.1. Let R be a ring and let $A = (A_{ij})_{i \in I}$ be 2-spherical. Then we associate a chamber system $\mathcal{C}^-(G_R^{\min})$ to $G_R^{\min}$ by taking the cosets $G_R^{\min}/\overline{B}_R^-$ as the set of chambers, and for any $i \in I$, letting $g\overline{B}_R^-$ be *i -adjacent* to $h\overline{B}_R^-$ if and only if $h^{-1}g \in \overline{P}_{iR}^-$. In particular,

$$g\overline{B}_R^- \sim_i h\overline{B}_R^- \iff \text{there exists } g_i \in \overline{G}_{iR}^- \text{ such that } h\overline{B}_R^- = gg_i\overline{B}_R^-.$$

We will write

$$(4.1) \quad gc_0^- := g\overline{B}_R^- \quad \text{for the chamber in } \mathcal{C}^-(G_R^{\min}) \text{ given by the coset } g\overline{B}_R^- \in G_R^{\min}/\overline{B}_R^-.$$

The group $G_R^{\min}$ acts on its chamber system by left multiplication, and this action preserves *i -adjacency* for any $i \in I$. In other words, $G_R^{\min}$ acts on $\mathcal{C}^-(G_R^{\min})$ by chamber automorphisms. Stabilisers of J -residues of the chamber c_0^- are given by the parabolic subgroups $\overline{P}_{JR}^-$. If $R = \mathbb{K}$ is a field, the chamber system $\mathcal{C}^-(G_{\mathbb{K}}) = \mathcal{C}^-(G_{\mathbb{K}}^{\min})$ is moreover a building (see [Mar18, Corollary 7.70]).

A chamber system is said to be *simply connected* (or more precisely, *simply 2-connected*) if every gallery which starts and ends in the same chamber c , is *2-homotopic* to the gallery (c) . Here a 2-homotopy consists of a finite sequence of *elementary 2-homotopies*, which in turn consist of replacing a J -subgallery (with $|J| \leq 2$) connecting two chambers by another J -subgallery connecting the same two chambers (see [DM07, §2] for the precise definition).

An important feature of the theory of groups acting on chamber systems, is the following.

Proposition 4.2 ([Tit86, Corollaire 1]). *Let $\mathcal{C} := (C, (\sim_i)_{i \in I})$ be a chamber system, and let G be a group acting transitively on C such that the action preserves i -adjacency. Let $c \in C$. Then the following are equivalent:*

- (a) $\mathcal{C}$ is simply connected;
- (b) G is the amalgamated product of the stabilisers

$$G_J := \text{Stab}_G(\mathcal{R}_J(c)) = \{g \in G \mid g \cdot c' \in \mathcal{R}_J(c), \text{ for all } c' \in \mathcal{R}_J(c)\},$$

for all $J \subseteq I$ with $|J| \leq 2$, along the canonical inclusions $G_{J_1} \subseteq G_{J_2}$ for $J_1 \subseteq J_2 \subseteq I$.

See e.g. [Ser80, §1] for a precise definition of such an amalgamated product. To obtain a presentation of a group G as an amalgamated product of a certain set of subgroups, it hence suffices to show that G acts on a simply connected chamber system and to identify the stabilisers of the rank 2 residues. To obtain chamber systems suited to our purposes, we need the following terminology.

Definition 4.3 ([DM07, §3]). Let $\mathcal{C} = (C, (\sim_i)_{i \in I})$ be a chamber system. A *filtration* of $\mathcal{C}$ is a sequence $\mathcal{F} := (C_n)_{n \in \mathbb{N}}$ of subsets of C , satisfying the following conditions:

- (i) $C_n \subseteq C_{n+1}$ for all $n \in \mathbb{N}$,
- (ii) $\bigcup_{n \in \mathbb{N}} C_n = C$,
- (iii) For every $n > 0$, if $C_{n-1} \neq \emptyset$, then there exists an $i \in I$ such that for all $c \in C_n$, there is some $c' \in C_{n-1}$ which is i -adjacent to c .

The filtration $\mathcal{F}$ is called *residual* if for every non-empty $J \subseteq I$ and any J -residue $\mathcal{R}_J$ in $\mathcal{C}$, the sequence $(C_n \cap \mathcal{R}_J)_{n \in \mathbb{N}}$ is a filtration of the chamber system $(\mathcal{R}_J, (\sim_j)_{j \in J})$. For any subset $X \subseteq C$, we define

$$(4.2) \quad \text{aff}(X) := C_{m_X} \cap X, \quad \text{where } m_X := \min\{n \in \mathbb{N} \mid C_n \cap X \neq \emptyset\}.$$

A residual filtration $\mathcal{F}$ is said to be *locally connected* if it satisfies the following condition:

- (lco) For any $J \subseteq I$ with $|J| \leq 2$, and any J -residue $\mathcal{R}_J$, the chamber system $(\text{aff}(\mathcal{R}_J), (\sim_j)_{j \in J})$ is connected.

Finally, $\mathcal{F}$ is said to be *locally simply connected* if it satisfies:

- (lsco) For any $J \subseteq I$ with $|J| \leq 3$, and any J -residue $\mathcal{R}_J$, the chamber system $(\text{aff}(\mathcal{R}_J), (\sim_j)_{j \in J})$ is simply connected.

We will make crucial use of the following three facts.

Proposition 4.4 ([DM07, Corollary 3.10 & Theorem 3.14]). *Let $\mathcal{F} = (C_n)_{n \in \mathbb{N}}$ be a residual filtration of a chamber system $\mathcal{C}$, with $C_0 \neq \emptyset$ and satisfying (lco) and (lsco). Then the following are equivalent:*

- (i) $\mathcal{C}$ is simply connected.
- (ii) C_0 is simply connected.

Proposition 4.5 ([Ron09, Theorem 4.3]). *Buildings are simply connected chamber systems.*

Proposition 4.6 ([MM25, Lemma 8.7]). *Let R be a local Bézout domain with field of fractions $\mathbb{K}$, satisfying (co) in case A is not spherical. Then the canonical map*

$$\mathcal{C}^-(G_R^{\min}) \rightarrow \mathcal{C}^-(G_{\mathbb{K}}), \quad g\overline{B}_R^- \rightarrow gB_{\mathbb{K}}^-$$

is an isomorphism of chamber systems.

In particular, in view of Proposition 4.5 and the fact that $\mathcal{C}^-(G_{\mathbb{K}})$ is a building by Example 4.1, under the hypotheses of Proposition 4.6, $\mathcal{C}^-(G_R^{\min})$ is simply connected. In the following section, we will construct a suitable filtration of $\mathcal{C}^-(G_R^{\min})$ such that $\overline{U}_R^+$ acts transitively on C_0 , and the latter will be simply connected by Proposition 4.4. We will then apply Proposition 4.2 to obtain Theorem B.

4.2. Filtration of $\mathcal{C}^-(G_R^{\min})$. We define the map

$$(4.3) \quad \delta: \mathcal{C}^-(G_R^{\min}) \rightarrow \mathcal{W}, \quad \text{via} \quad \delta(gc_0^-) = w \iff g \in B(w).$$

This is well-defined by (3.2) and the definition of $B(w)$ (see (3.1)). As in section 2, let ℓ denote the length function on $\mathcal{W}$, with respect to $\{s_i \mid i \in I\}$.

Lemma 4.7. *Let R be a local ring, with residue field k . Let $J \subseteq I$ and let $\mathcal{R}_J$ be a J -residue in $\mathcal{C}^-(G_R^{\min})$. Let $gc_0^- \in \mathcal{R}_J$ with $\delta(gc_0^-) = w$. Set moreover*

$$l := \min \{ \ell(\delta(hc_0^-)) \mid hc_0^- \in \mathcal{R}_J \}.$$

Then the following hold:

(a) *For any $hc_0^- \in \mathcal{R}_J$, one has $\delta(hc_0^-) \in w\mathcal{W}_J$. In particular,*

$$(4.4) \quad \{ \delta(hc_0^-) \mid hc_0^- \in \mathcal{R}_J \} = w\mathcal{W}_J.$$

(b) *$\text{aff}(\mathcal{R}_J) = \{ hc_0^- \in \mathcal{R}_J \mid \ell(\delta(hc_0^-)) = l \} = \{ hc_0^- \in \mathcal{R}_J \mid \delta(hc_0^-) = w_J \}$, where w_J is the unique element of $w\mathcal{W}_J$ of minimal length.*

Proof. Let gc_0^- be as in the statement. Let moreover $hc_0^- \in \mathcal{R}_J$. Let $(i_1, \dots, i_d)$ be the type of a J -gallery from gc_0^- to hc_0^- . Then for $1 \leq t \leq d$, there exist $g_t \in \overline{G}_{i_t R}$ such that

$$gc_0^- \sim_{j_1} gg_1 c_0^- \sim_{j_2} \dots \sim_{j_d} gg_1 \dots g_d c_0^- = hc_0^-.$$

Now, $p(g) \in U_k^+ \tilde{w} B_k^-$ by Proposition 3.2. Moreover, since $g_t \in \overline{G}_{i_t R}$ for all $1 \leq t \leq d$, we have $p(g_t) \in B_k^- \sqcup B_k^- \tilde{s}_{i_t} B_k^-$. Hence by (2.4), there is some subword $(s_{j_1}, \dots, s_{j_n})$ of $(s_{i_1}, \dots, s_{i_d}) \in \{s_j \mid j \in J\}^d$ such that

$$p(g_1 \dots g_d) \in B_k^- \tilde{s}_{j_1} \dots \tilde{s}_{j_n} B_k^-.$$

It follows that

$$p(h) \in U_k^+ \tilde{w} B_k^- \tilde{s}_{j_1} \dots \tilde{s}_{j_n} B_k^-,$$

and hence by (2.5), there is some subword $\mathbf{w}'$ of $(s_{j_1}, \dots, s_{j_n})$ such that for the associated element $w' \in \mathcal{W}_J$, we have $p(h) \in U_k^+ \tilde{w} \tilde{w}' B_k^-$. By (3.3) it now follows that $h \in B(w w')$. In particular, $\delta(hc_0^-) \in w\mathcal{W}_J$. This shows one inclusion of (4.4).

Conversely, by definition, every element $hc_0^- \in \mathcal{R}_J$ is of the form gxc_0^- for some $x \in \overline{P}_{JR}^-$. Since $p(g) \in U_k^+ \tilde{w} B_k^-$, there exist $u_+ \in \overline{U}_R^+$ and $b_- \in \overline{B}_R^-$ such that $p(g) = p(u_+ \tilde{w} b_-)$. Then, for any $w' \in \mathcal{W}_J$, setting $x := b_-^{-1} \tilde{w}' \in \overline{P}_{JR}^-$, we have that

$$p(g)p(x) = p(u_+ \tilde{w} \tilde{w}') \in U_k^+ \tilde{w} \tilde{w}' B_k^-.$$

In other words, for any $w' \in \mathcal{W}_J$, there is a chamber $hc_0^- = gxc_0^- \in \mathcal{R}_J$ such that $\delta(hc_0^-) = ww'$, which yields the other inclusion. This shows (a). Now (b) follows from (a) and the fact that w_J is the unique element of minimal length in $w\mathcal{W}_J$. $\square$

Fix an injection $|\cdot|: \mathcal{W} \rightarrow \mathbb{N}$ satisfying:

- (i) $|1_{\mathcal{W}}| = 0$,
- (ii) $\ell(w_1) < \ell(w_2) \implies |w_1| < |w_2|$.

Proposition 4.8. *Let R be a local ring with residue field k . Let for any $n \in \mathbb{N}$,*

$$(4.5) \quad C_n := \{ gc_0^- \in \mathcal{C}^-(G_R^{\min}) \mid |\delta(gc_0^-)| \leq n \}.$$

Then $(C_n)_{n \geq 0}$ is a residual filtration of $\mathcal{C}^-(G_R^{\min})$.

Proof. Let $\mathcal{R}_J$ be a J -residue in $\mathcal{C}^-(G_R^{\min})$. It is clear that for any $n \in \mathbb{N}$, $\mathcal{R}_J \cap C_n \subseteq \mathcal{R}_J \cap C_{n+1}$. Moreover, $\bigcup_{n \in \mathbb{N}} (\mathcal{R}_J \cap C_n) = \mathcal{R}_J$.

All that remains to show is that for every $n > 0$ such that $C_{n-1} \cap \mathcal{R}_J \neq \emptyset$, there exists a $j \in J$ such that every chamber $c \in C_n \cap \mathcal{R}_J$ is j -adjacent to some chamber $c' \in C_{n-1} \cap \mathcal{R}_J$. There are two possible cases: either $C_{n-1} \cap \mathcal{R}_J = C_n \cap \mathcal{R}_J$, or there exists an element $gc_0^- \in (C_n \setminus C_{n-1}) \cap \mathcal{R}_J$. In the former case, we may choose $j \in J$ arbitrarily, since every chamber $c \in C_n \cap \mathcal{R}_J = C_{n-1} \cap \mathcal{R}_J$ is j -adjacent to itself. In the latter case, by definition there is a fixed $w \in \mathcal{W}$ such that $\delta(gc_0^-) = w$, for all $gc_0^- \in (C_n \setminus C_{n-1}) \cap \mathcal{R}_J$. Moreover, $w\mathcal{W}_J$ has a unique element of minimal length w_J , and since $C_{n-1} \cap \mathcal{R}_J$ is non-empty, this element is different from

w . Indeed, define $m = \min\{r \in \mathbb{N} \mid C_r \cap \mathcal{R}_J \neq \emptyset\}$. Then $m < n$, and with l as in Lemma 4.7, i.e. $l = \ell(w_J)$,

$$\begin{aligned} C_m \cap \mathcal{R}_J &= \{gc_0^- \in \mathcal{R}_J \mid |gc_0^-| = m\} = \{gc_0^- \in \mathcal{R}_J \mid \ell(\delta(gc_0^-)) = l\} \\ &= \{gc_0^- \in \mathcal{R}_J \mid \delta(gc_0^-) = w_J\}. \end{aligned}$$

In particular, by Lemma 4.7,

$$w = w_J w', \text{ with } w' \in \mathcal{W}_J \setminus \{1_{\mathcal{W}}\}.$$

Let now $j \in J$ such that $\ell(w' s_j) = \ell(w') - 1$, and let moreover $gc_0^- \in (C_n \setminus C_{n-1}) \cap \mathcal{R}_J$, i.e. $\delta(gc_0^-) = w$. Choose $y \in \varphi_{jR}(\mathrm{SL}_2(R) - U_2^+(L)B_2^-(R))$. We show that $\delta(gyc_0^-) = ws_j$. Indeed, by Lemma 2.7, we have that $p(y) \notin \varphi_{jk}(B_2^-(k))$. In particular, $p(y) \in \varphi_{ik}(B_2^-(k)) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} B_2^-(k)$, and hence $p(y) \in B_k^- \tilde{s}_j B_k^-$. It follows that

$$p(gy) \in B_k^+ \tilde{w} B_k^- \tilde{s}_j B_k^- = B_k^+ \tilde{w} \tilde{s}_j B_k^-,$$

where the latter equality follows by (2.5), since

$$\ell(ws_j) = \ell(w_J w' s_j) = \ell(w_J) + \ell(w' s_j) = \ell(w_J) + \ell(w') - 1 = \ell(w) - 1.$$

Hence $gy \in B(ws_j)$ by (3.3), i.e. $\delta(gyc_0^-) = ws_j$. In particular,

$$\ell(\delta(gyc_0^-)) < \ell(\delta(gc_0^-))$$

and by definition of $|\cdot|: \mathcal{W} \rightarrow \mathbb{N}$, it follows that $gyc_0^- \in C_{n-1}$. By construction $gyc_0^- \in \mathcal{R}_J$. Hence $j \in J$ is an element such that every $gc_0^- \in C_n \cap \mathcal{R}_J$ is j -adjacent to some $hc_0^- \in C_{n-1} \cap \mathcal{R}_J$, finishing the proof. $\square$

Next, we show the condition (lco). We calculate what the set $\mathrm{aff}(\mathcal{R}_J)$ looks like for a residue $\mathcal{R}_J$ in $\mathcal{C}^-(G_R^{\min})$.

Proposition 4.9. *Let $\mathcal{R}_J$ be a J -residue in $\mathcal{C}^-(G_R^{\min})$, with $J \subseteq I$. Let $hc_0^- \in \mathcal{R}_J$ be such that $\delta(hc_0^-) = w$, and let w_J be the unique element of minimal length in $w\mathcal{W}_J$. Then there is some $gc_0^- \in \mathcal{R}_J$ such that $\delta(gc_0^-) = w_J$, and*

$$(4.6) \quad \mathrm{aff}(\mathcal{R}_J) = \left\{ gxc_0^- \in \mathcal{C}^-(G_R^{\min}) \mid x \in \overline{U}_{JR}^+ \right\}.$$

Proof. Note that a chamber $gc_0^- \in \mathcal{R}_J$ with $\delta(gc_0^-) = w_J$ exists by Lemma 4.7(a). Then $p(g) \in B_k^+ \tilde{w}_J B_k^-$, and in particular, there exists $b_+ \in \overline{B}_R^+$ such that $p(g)B_k^- = p(b_+) \tilde{w}_J B_k^-$. Hence $b_+ \tilde{w}_J$ is a representative of the coset $g\overline{B}_R^-$, i.e. $gc_0^- = b_+ \tilde{w}_J c_0^-$. We then have

$$\mathcal{R}_J = \left\{ b_+ \tilde{w}_J x c_0^- \in \mathcal{C}^-(G_R^{\min}) \mid x \in \overline{G}_{JR} \right\}.$$

Now, by Lemma 4.7(b),

$$b_+ \tilde{w}_J x c_0^- \in \mathrm{aff}(\mathcal{R}_J) \iff p(b_+ \tilde{w}_J x) \in B_k^+ \tilde{w}_J B_k^-.$$

Moreover, G_{Jk} has a Birkhoff decomposition $G_{Jk} = \coprod_{w' \in \mathcal{W}_J} U_{Jk}^+ \tilde{w}' B_{Jk}^-$. In particular, for any $x \in \overline{G}_{JR}$, there is a $w' \in \mathcal{W}_J$ such that $p(x) \in U_{Jk}^+ \tilde{w}' B_{Jk}^-$, and hence

$$p(b_+ \tilde{w}_J x) \in B_k^+ \tilde{w}_J U_{Jk}^+ \tilde{w}' B_{Jk}^-.$$

By Lemma 2.4, since w_J is right J -reduced, we have that $\tilde{w}_J U_{Jk}^+ \subseteq B_k^+ \tilde{w}_J$. In particular,

$$p(b_+ \tilde{w}_J x) \in B_k^+ \tilde{w}_J U_{Jk}^+ \tilde{w}' B_{Jk}^- \subseteq B_k^+ \tilde{w}_J \tilde{w}' B_{Jk}^-.$$

Hence $b_+ \tilde{w}_J x c_0^- \in \mathrm{aff}(\mathcal{R}_J)$ if and only if $w' = 1$, i.e. $p(x) \in U_{Jk}^+ B_{Jk}^-$. By Lemma 2.5, it follows that $b_+ \tilde{w}_J x c_0^- \in \mathrm{aff}(\mathcal{R}_J)$ if and only if $x \in \overline{U}_{JR}^+ \overline{U}_L^+ \overline{B}_R^- \cap \overline{G}_{JR}$. Now,

$$\begin{aligned} \overline{U}_{JR}^+ \overline{U}_L^+ \overline{B}_R^- \cap \overline{G}_{JR} &= \overline{U}_{JR}^+ \left(\overline{U}_L^+ \overline{B}_R^- \cap \overline{G}_{JR} \right) \subseteq \overline{U}_{JR}^+ \left(\overline{B}_R^+ \overline{B}_R^- \cap \overline{G}_{JR} \right) \\ &\subseteq \overline{U}_{JR}^+ \left(\overline{U}_{JR}^+ \overline{U}_{JR}^- T_R \right) \subseteq \overline{U}_{JR}^+ \overline{B}_R^-, \end{aligned}$$

where the inclusion $\overline{B}_R^+ \overline{B}_R^- \cap \overline{G}_{JR} \subseteq \overline{U}_{JR}^+ \overline{U}_{JR}^- T_R$ follows from Lemma 2.6. It follows in particular that $b_+ \tilde{w}_J x c_0^- \in \text{aff}(\mathcal{R}_J)$ if and only if $x \in \overline{U}_{JR}^+ \overline{B}_R^-$, showing the claim. $\square$

Proposition 4.10. *Let A be 2-spherical, and R a local ring satisfying (co). Then the filtration $(C_n)_{n \geq 0}$ of $\mathcal{C}^-(G_R^{\min})$ as defined in (4.5) satisfies (lco).*

Proof. By Proposition 4.9, for any J -residue $\mathcal{R}_J$, there is some $g \in G_R^{\min}$ such that any two elements in $\text{aff}(\mathcal{R}_J)$ are of the form gxc_0^- and gyc_0^- , with $x, y \in \overline{U}_{JR}^+$. Since R is assumed to satisfy (co), it follows by Lemma 2.3 that $x^{-1}y$ can be expressed as a product of elements in the simple root subgroups $U_{\alpha_j R}$ for $j \in J$. Hence there exists a J -gallery from gxc_0^- to gyc_0^- , which is moreover completely contained in $\text{aff}(\mathcal{R}_J)$ by Proposition 4.9. $\square$

We now prove Theorem B.

Theorem 4.11. *Let $A = (A_{ij})_{i,j \in I}$ be a 2-spherical GCM and R a local Bézout domain satisfying (co). Then $\overline{U}_R^+$ is the amalgamated product of its subgroups $\overline{U}_{JR}^+$ for $J \subseteq I$ and $|J| \leq 2$, along the canonical inclusions $\overline{U}_{J_1 R}^+ \hookrightarrow \overline{U}_{J_2 R}^+$ for all $J_1 \subseteq J_2$, if this is the case for all of its rank 3 subgroups $\overline{U}_{I' R}^+$, i.e. for all $I' \subseteq I$ with $|I'| = 3$.*

Proof. Under the hypotheses on A and R , the chamber system $\mathcal{C}^-(G_R^{\min})$ is a building by Proposition 4.6. In particular, $\mathcal{C}^-(G_R^{\min})$ is simply connected by Proposition 4.5. The filtration $(C_n)_{n \in \mathbb{N}}$ of $\mathcal{C}^-(G_R^{\min})$ as defined in (4.5), is a residual filtration satisfying (lco) by Proposition 4.8 and Proposition 4.10. Moreover, by definition, $C_0 = \overline{U}_R^+ c_0^-$. Hence if $(C_n)_{n \in \mathbb{N}}$ were to satisfy (lsco), then by Proposition 4.4, $\overline{U}_R^+ c_0^-$ would be a simply connected chamber system, on which $\overline{U}_R^+$ acts transitively.

In particular, if $(C_n)_{n \in \mathbb{N}}$ satisfies (lsco), we find that $\overline{U}_R^+$ has the desired amalgamated product decomposition by Proposition 4.2, since the stabilisers in $\overline{U}_R^+$ of the residues of rank at most 2 containing c_0^- are the subgroups $\overline{U}_R^+ \cap \overline{P}_{JR}^- = \overline{U}_{JR}^+$ (see (2.9)) with $|J| \leq 2$, by Example 4.1.

By definition, $(C_n)_{n \in \mathbb{N}}$ satisfies (lsco) if and only if for any J -residue $\mathcal{R}_J \subseteq \mathcal{C}^-(G_R^{\min})$ with $|J| = 3$, the set $\text{aff}(\mathcal{R}_J)$ is simply connected. By Proposition 4.9, this is equivalent to the set $g\overline{U}_{JR}^+ c_0^-$ being simply connected, for all $g \in G_R^{\min}$ and all $J \subseteq I$ of size 3. But since the action of $G_R^{\min}$ on $\mathcal{C}^-(G_R^{\min})$ preserves i -adjacency for any $i \in I$, we have that for any $g \in G_R^{\min}$ and any $J \subseteq I$,

$$g\overline{U}_{JR}^+ c_0^- \cong \overline{U}_{JR}^+ c_0^-,$$

as chamber systems. In combination with Proposition 4.2, it follows that $(C_n)_{n \in \mathbb{N}}$ satisfies (lsco) if and only if for any $J \subseteq I$ with $|J| = 3$, the subgroup $\overline{U}_{JR}^+ \leq \overline{U}_R^+$ is the amalgamated product of its subgroups $\overline{U}_{JR}^+ \cap \overline{P}_{J'R}^- = \overline{U}_{J'R}^+$ (see (2.9)) with $J' \subseteq J$ and $|J'| \leq 2$, along the inclusions $\overline{U}_{J_1 R}^+ \subseteq \overline{U}_{J_2 R}^+$ for all $J_1 \subseteq J_2 \subseteq J$. $\square$

5. SIMPLE CONNECTEDNESS IN TYPE A_3

5.1. The reduction step and proof in the A_3 case. In [Tit86], Tits showed that for a Cartan matrix A , the problem of obtaining a presentation of U_R^+ as an amalgamated product of its rank 2 subgroups, reduces to its rank 3 subgroups having such a presentation. He then moreover showed that for a Cartan matrix A of type A_3 , and $\mathbb{K}$ a field different from $\mathbb{F}_2$, the group $U_{\mathbb{K}}^+$ is the amalgam of its rank 2 subgroups. These result were then combined to obtain an amalgamated product decomposition of $U_{\mathbb{K}}^+$ in terms of its rank 2 subgroups for any Cartan matrix of simply laced type.

We adapt Tits' strategy to the more general setting of commutative rings satisfying a minor condition which extends the requirement that $\mathbb{K} \neq \mathbb{F}_2$, proving Theorem A.

First, we formulate the statement of Tits' reduction step to the rank 3 case for spherical type GCM's, for the convenience of the reader.

Proposition 5.1 ([Tit86, Proposition 7]). *Let $A = (A_{ij})_{i,j \in I}$ be a Cartan matrix, with simple roots $\{\alpha_i \mid i \in I\}$. Let R be a (commutative, unital) ring. Suppose the following two conditions are satisfied for $U_R^+ \leq \mathfrak{G}_A(R)$:*

- (i) *For any distinct $i, j \in I$, the group $U_{\{i,j\}R}^+$ is generated by $U_{\alpha_i R}$ and $U_{\alpha_j R}$;*
- (ii) *For any distinct $i, j, k \in I$, the group $U_{\{i,j,k\}R}^+$ is the amalgamated product of its subgroups U_{JR}^+ with $J \subseteq \{i, j, l\}$ and $|J| \leq 2$, along the canonical inclusions of the subgroups $U_{\alpha_i R}$, $U_{\alpha_j R}$ and $U_{\alpha_l R}$.*

Then U_R^+ itself is the amalgamated product of its subgroups U_{JR}^+ for all $J \subseteq I$ with $|J| \leq 2$, along the canonical inclusions of the root subgroups $U_{\alpha_i R} \hookrightarrow U_{JR}^+$ for all $i \in J \subseteq I$.

Remark 5.2. The condition (i) holds as soon as R satisfies (co), by [All16, Lemma 11.1]. In particular, for a simply laced Cartan matrix, the condition (i) is satisfied for any commutative unital ring R .

For the rest of this section, let A be of type A_3 . Note that in this case, for any R ,

$$G_R = E_4(R) \leq \mathrm{SL}_4(R), \quad \text{and} \quad U_R^+ = \left\{ \begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 1 \end{pmatrix} \in E_4(R) \right\}.$$

Here $E_4(R)$ denotes the subgroup of $\mathrm{SL}_4(R)$ generated by the elementary matrices $I_4 + \lambda E_{i,j}$, for all $\lambda \in R$, where I_4 denotes the identity matrix and $E_{i,j}$ denotes the matrix which has every entry equal to 0, except the entry (i, j) , which equals 1. For any $i \in I = \{1, 2, 3\}$, the root subgroup U_{α_i} of U_R^+ is generated by the elements $I_4 + \lambda E_{i,i+1}$, for all $\lambda \in R$.

Theorem 5.3. *Let R be a (commutative, unital) ring such that 1_R is the sum of two invertible elements. Let $A = (A_{ij})_{i,j \in I}$ be the Cartan matrix of type A_3 . Then $U^+ := U_R^+ \leq G_R$ is the amalgamated product of the simple root subgroups U_{α_i} and the rank two subgroups $U_{JR}^+ = \langle U_{\alpha_i}, U_{\alpha_j} \rangle$, along the canonical inclusions $U_{\alpha_i} \hookrightarrow U_{JR}^+$, for all $J = \{i, j\} \subseteq I$.*

When R is a field, this statement is precisely [Tit86, Proposition 8], which we formulate here for completeness' sake.

Proposition 5.4 ([Tit86, Proposition 8]). *Let $\mathbb{K}$ be a field different from $\mathbb{F}_2$ and let $A = (A_{ij})_{i,j \in I}$ be the Cartan matrix of type A_3 . Then $U_{\mathbb{K}}^+ \leq G_{\mathbb{K}} = \mathrm{SL}_4(\mathbb{K})$ is the amalgamated product of the simple root subgroups U_{α_i} and the rank two subgroups $U_{J\mathbb{K}}^+ = \langle U_{\alpha_i}, U_{\alpha_j} \rangle$, along the canonical inclusions $U_{\alpha_i} \hookrightarrow U_{J\mathbb{K}}^+$, for all $J = \{i, j\} \subseteq I$.*

Before we proceed with the proof of Theorem 5.3, we first introduce some notation. By definition, $U^+ := U_R^+$ is the group with generators

$$x_{\alpha}(r), \text{ for all } \alpha \in \Delta := \{\alpha_1, \alpha_2, \alpha_3, \alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_1 + \alpha_2 + \alpha_3\} \text{ and all } r \in R,$$

and relations, for all $r, s \in R$:

- (i) $x_{\alpha}(r)x_{\alpha}(s) = x_{\alpha}(r+s)$, for all $\alpha \in \Delta$,
- (ii) $[x_{\alpha_1}(r), x_{\alpha_2}(s)] = x_{\alpha_1+\alpha_2}(rs)$, $[x_{\alpha_1+\alpha_2}(r), x_{\alpha_1}(s)] = [x_{\alpha_1+\alpha_2}(r), x_{\alpha_2}(s)] = 1$,
- (iii) $[x_{\alpha_2}(r), x_{\alpha_3}(s)] = x_{\alpha_2+\alpha_3}(rs)$, $[x_{\alpha_2+\alpha_3}(r), x_{\alpha_2}(s)] = [x_{\alpha_2+\alpha_3}(r), x_{\alpha_3}(s)] = 1$,
- (iv) $[x_{\alpha_1}(r), x_{\alpha_3}(s)] = 1$
- (v) $[x_{\alpha_1+\alpha_2}(r), x_{\alpha_2+\alpha_3}(s)] = 1$
- (vi) $[x_{\alpha_1+\alpha_2}(r), x_{\alpha_3}(s)] = [x_{\alpha_1}(r), x_{\alpha_2+\alpha_3}(s)]$
- (vii) $[x_{\alpha_1+\alpha_2}(r), x_{\alpha_3}(s)] = x_{\alpha_1+\alpha_2+\alpha_3}(rs)$,
- (viii) $[x_{\alpha_1+\alpha_2+\alpha_3}(r), x_{\alpha}(s)] = 1$, for all $\alpha \in \Delta$.

Moreover, the amalgamated product $\tilde{U}$ of the root subgroups U_{α_i} and the rank two subgroups $U_{i,j} := U_{\{i,j\}R}^+$ of U^+ along the inclusions of U_{α_i} into $U_{i,j}$, is given by the group with the generators $x_{\alpha}(r)$, for all $r \in R$ and all $\alpha \in \tilde{\Delta} := \Delta \setminus \{\alpha_1 + \alpha_2 + \alpha_3\}$, satisfying the relations (i) for all $\alpha \in \tilde{\Delta}$, as well as the relations (ii), (iii) and (iv).

Define in $\tilde{U}$, for any $t \in R$,

$$(5.1) \quad v_{123}(t) := [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(t)].$$

Proving Theorem 5.3 consists of showing that $\tilde{U}$ also satisfies eqs. (v) to (viii), with the elements $x_{\alpha_1+\alpha_2+\alpha_3}(t)$ replaced by $v_{123}(t)$, and that $v_{123}(t)$ also satisfies (i).

Let $U_i := U_{\alpha_i}$ for all $i \in I$. Moreover, let V_i be the conjugation of U_i by $x_{\alpha_2}(1)$, i.e. $V_i := x_{\alpha_2}(-1)U_i x_{\alpha_2}(1)$. In particular, by (ii) and (iii),

$$(5.2) \quad V_1 = \langle x_{\alpha_1}(r)x_{\alpha_1+\alpha_2}(r) \mid r \in R \rangle$$

$$(5.3) \quad V_3 = \langle x_{\alpha_3}(r)x_{\alpha_2+\alpha_3}(-r) \mid r \in R \rangle.$$

Remark 5.5. As Tits remarked in [Tit86, §16.5], Proposition 5.4 does not hold over $\mathbb{F}_2$. Indeed, if one supposes $\mathbb{K} = \mathbb{F}_2$, then we define, for $i \in \{1, 3\}$, the groups $D_i := U_i * V_i$, which are isomorphic to the infinite dihedral group D_{∞} . Moreover, we can define the semidirect product

$$H := (D_1 \times D_3) \rtimes U_2,$$

by letting the elements $x_{\alpha_2}(1)$ act on D_i by conjugation (respecting the relations (ii) and (iii)). Then the relations (i), (ii), (iii) and (iv) are satisfied in H , and in particular $H \cong \tilde{U}$. It follows that $\tilde{U} \not\cong U$, since U is the finite group

$$U \cong \left\{ \begin{pmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \mathrm{SL}_4(\mathbb{F}_2) \right\}.$$

The following lemma is an intermediary step towards the proof of Theorem 5.3.

Lemma 5.6 ([Tit86, §16 (7)]). *For U_i and V_i as above, the following commutation relations hold in $\tilde{U}$:*

$$[U_1, U_3] = [U_3, V_3] = [V_1, V_3] = [U_1, V_1] = \{1\}.$$

Proof. The groups U_1 and U_3 commute by (iv), and conjugating relation (iv) by $x_{\alpha_2}(1)$ shows that V_1 and V_3 commute. Now U_1 commutes with V_1 by (5.2) and (ii), and U_3 commutes with V_3 by (5.3) and (iii). $\square$

Moreover, the following holds.

Lemma 5.7 ([Tit86, Lemme 4]). *Let $u_i \in U_i$ and $v_i \in V_i$ be such that $[u_1 v_1, u_3 v_3] = 1$. Then*

$$[u_1, v_3] = [u_3, v_1],$$

and $[u_1, v_3]$ is central in the subgroup X of $\tilde{U}$ generated by U_1, U_3, V_1 and V_3 .

Proof. Suppose $u_1 v_1 u_3 v_3 = u_3 v_3 u_1 v_1$. Then $v_1 u_3 v_3 = u_1^{-1} u_3 v_3 u_1 v_1$. But $v_1 u_3 v_3 = v_3 v_1 u_3$ since v_3 commutes with u_3 and v_3 by Lemma 5.6. Hence

$$v_1 u_3 = v_3^{-1} u_1^{-1} u_3 v_3 u_1 v_1.$$

But the latter is equal to $u_3 v_3^{-1} u_1^{-1} v_3 u_1 v_1$, since u_3 commutes with u_1 and v_3 . Hence

$$u_3^{-1} v_1 u_3 = v_3^{-1} u_1^{-1} v_3 u_1 v_1 = v_1 v_3^{-1} u_1^{-1} v_3 v_1,$$

where the last equality follows from the fact that v_1 commutes with u_1 and v_3 . Hence

$$v_1^{-1} u_3^{-1} v_1 u_3 = v_3^{-1} u_1^{-1} v_3 u_1.$$

This shows the first statement. It now trivially follows that $[u_1, v_3]$ is central in X , since, by Lemma 5.6, $[u_1, v_3]$ commutes with U_3 and V_1 , and $[u_3, v_1]$ commutes with U_1 and V_3 . $\square$

Proof of Theorem 5.3. Consider the group $\tilde{U}$ as defined above. We show the relations eqs. (v) to (viii) hold, with the elements $x_{\alpha_1+\alpha_2+\alpha_3}(t)$ replaced by $v_{123}(t)$, and that $v_{123}(t)$ also satisfies (i). We begin by showing relation (vi). Let $r, s \in R$. Then by (ii),

$$[x_{\alpha_1}(r), x_{\alpha_3}(s)] = 1.$$

Conjugating this expression by $x_{\alpha_2}(t)$ for $t \in R$ implies that

$$[x_{\alpha_1}(r)x_{\alpha_1+\alpha_2}(rt), x_{\alpha_3}(s)x_{\alpha_2+\alpha_3}(-st)] = 1.$$

In particular, by (i),

$$[x_{\alpha_1}(r - rt)x_{\alpha_1}(rt)x_{\alpha_1+\alpha_2}(rt), x_{\alpha_3}(s - st)x_{\alpha_3}(st)x_{\alpha_2+\alpha_3}(-st)] = 1.$$

But by (5.2) and (5.3),

$$\begin{aligned} x_{\alpha_1}(r - rt) &\in U_1, & x_{\alpha_1}(rt)x_{\alpha_1+\alpha_2}(rt) &\in V_1, \text{ and} \\ x_{\alpha_3}(s - st) &\in U_3, & x_{\alpha_3}(st)x_{\alpha_2+\alpha_3}(-st) &\in V_3. \end{aligned}$$

Hence we may apply Lemma 5.7 to obtain that for all $r, s, t \in R$

$$(5.4) \quad [x_{\alpha_1}(r - rt), x_{\alpha_3}(st)x_{\alpha_2+\alpha_3}(-st)] = [x_{\alpha_3}(s - st), x_{\alpha_1}(rt)x_{\alpha_1+\alpha_2}(rt)],$$

and moreover these elements are central in the subgroup X of $\tilde{U}$ generated by U_1, U_3, V_1 and V_3 . In any group G , the following relations hold for $x, y, z \in G$:

$$(5.5) \quad [x, zy] = [x, y][x, z]^y$$

$$(5.6) \quad [xz, y] = [x, y]^z[z, y],$$

It follows that

$$\begin{aligned} [x_{\alpha_1}(r - rt), x_{\alpha_3}(st)x_{\alpha_2+\alpha_3}(-st)] &= [x_{\alpha_1}(r - rt), x_{\alpha_2+\alpha_3}(-st)] [x_{\alpha_1}(r - rt), x_{\alpha_3}(st)]^{x_{\alpha_2+\alpha_3}(-st)} \\ &= [x_{\alpha_1}(r - rt), x_{\alpha_2+\alpha_3}(-st)], \end{aligned}$$

where the last equality follows from (iv). Similarly,

$$[x_{\alpha_3}(s - st), x_{\alpha_1}(rt)x_{\alpha_1+\alpha_2}(rt)] = [x_{\alpha_3}(s - st), x_{\alpha_1+\alpha_2}(rt)].$$

However, in any group G there are moreover the relations

$$(5.7) \quad [x^{-1}, y] = [y, x]^{x^{-1}} \text{ and } [x, y^{-1}] = [y, x]^{y^{-1}}, \quad \text{for all } x, y, z \in G.$$

Applying (5.7) to the previous expression, we get

$$\begin{aligned} [x_{\alpha_3}(s - st), x_{\alpha_1+\alpha_2}(rt)] &= [x_{\alpha_1+\alpha_2}(rt), x_{\alpha_3}(-s + st)]^{x_{\alpha_1+\alpha_2}(s-st)} \\ &= [x_{\alpha_1+\alpha_2}(rt), x_{\alpha_3}(-s + st)], \end{aligned}$$

where the last equality holds because $[x_{\alpha_1+\alpha_2}(rt), x_{\alpha_3}(-s + st)]$ is central in X , and because $x_{\alpha_1+\alpha_2}(s - st) \in X$. We conclude from (5.4) that

$$(5.8) \quad \forall r, s, t \in R, \quad [x_{\alpha_1}(r - rt), x_{\alpha_2+\alpha_3}(-st)] = [x_{\alpha_1+\alpha_2}(rt), x_{\alpha_3}(-s + st)].$$

In particular,

$$(5.9) \quad \text{If there exists a } t \in R \text{ such that } t, 1 - t \in R^\times, \text{ then for any } r, s \in R, \text{ the elements } [x_{\alpha_1}(r), x_{\alpha_2+\alpha_3}(s)] \text{ and } [x_{\alpha_1+\alpha_2}(r), x_{\alpha_3}(s)] \text{ are central in } X.$$

We now apply a change of variables to the above. Let $\lambda \in R^\times$ such that $1 + \lambda$ is invertible. Take $t := \lambda(1 + \lambda)^{-1}$. Then t and $1 - t = (\lambda + 1)^{-1}$ are invertible. In particular, by (5.8),

$$(5.10) \quad \text{For any } \lambda \in R \text{ such that } \lambda, \lambda + 1 \in R^\times, \text{ and for any } r, s \in R, \\ [x_{\alpha_1}(r), x_{\alpha_2+\alpha_3}(s)] = [x_{\alpha_1+\alpha_2}(r\lambda), x_{\alpha_3}(s\lambda^{-1})].$$

By assumption, there exists an element $b \in R^\times$ such that $1 + b \in R^\times$. Then $1 + (-b - 1) \in R^\times$, and $1 - b^{-1}(b + 1) = b^{-1} \in R^\times$. In particular by (5.10), one has that, for any $r, s \in R$,

$$\begin{aligned} [x_{\alpha_1+\alpha_2}(r), x_{\alpha_3}(s)] &= [x_{\alpha_1}(rb^{-1}), x_{\alpha_2+\alpha_3}(sb)] \\ &= [x_{\alpha_1+\alpha_2}(-rb^{-1}(b+1)), x_{\alpha_3}(-sb(b+1)^{-1})] \\ &= [x_{\alpha_1}(r), x_{\alpha_2+\alpha_3}(s)], \end{aligned}$$

and relation (vi) follows.

Next, we show (v). By definition, for any $r, s \in R$,

$$\begin{aligned} [x_{\alpha_1+\alpha_2}(r), x_{\alpha_2+\alpha_3}(s)] &= [[x_{\alpha_1}(r), x_{\alpha_2}(1)], [x_{\alpha_3}(-s), x_{\alpha_2}(1)]] \\ &= [x_{\alpha_1}(-r)x_{\alpha_2}(-1)x_{\alpha_1}(r)x_{\alpha_2}(1), x_{\alpha_3}(s)x_{\alpha_2}(-1)x_{\alpha_3}(-s)x_{\alpha_2}(1)]. \end{aligned}$$

Denote

$$\begin{aligned} u_1 &:= x_{\alpha_1}(-r) \in U_1, & v_1 &:= x_{\alpha_2}(-1)x_{\alpha_1}(r)x_{\alpha_2}(1) \in V_1, \\ u_3 &:= x_{\alpha_3}(s) \in U_3, & v_3 &:= x_{\alpha_2}(-1)x_{\alpha_3}(-s)x_{\alpha_2}(1) \in V_3. \end{aligned}$$

Then we obtain by (5.6) that

$$\begin{aligned} [x_{\alpha_1+\alpha_2}(r), x_{\alpha_2+\alpha_3}(s)] &= [u_1, u_3v_3]^{v_1}[v_1, u_3v_3] = [u_1, v_3]^{v_1}[u_1, u_3]^{v_3v_1}[v_1, u_3v_3] \\ &= [u_1, v_3][v_1, u_3v_3], \end{aligned}$$

where the last equality holds because u_1 commutes with u_3 , and v_1 commutes with u_1 and v_3 by Lemma 5.6. Applying (5.5) we then obtain that

$$\begin{aligned} [x_{\alpha_1+\alpha_2}(r), x_{\alpha_2+\alpha_3}(s)] &= [u_1, v_3][v_1, v_3][v_1, u_3]^{v_3} \\ &= [u_1, v_3][v_1, u_3], \end{aligned}$$

where the latter equality follows from Lemma 5.6. In particular, by (ii) and (iii),

$$\begin{aligned} [x_{\alpha_1+\alpha_2}(r), x_{\alpha_2+\alpha_3}(s)] &= [x_{\alpha_1}(-r), x_{\alpha_2}(-1)x_{\alpha_3}(-s)x_{\alpha_2}(1)][x_{\alpha_2}(-1)x_{\alpha_1}(r)x_{\alpha_2}(1), x_{\alpha_3}(s)] \\ &= [x_{\alpha_1}(-r), x_{\alpha_3}(-s)x_{\alpha_2+\alpha_3}(s)][x_{\alpha_1}(r)x_{\alpha_1+\alpha_2}(r), x_{\alpha_3}(s)]. \end{aligned}$$

Applying once more (5.5) and (5.6), as well as Lemma 5.6, yields

$$[x_{\alpha_1+\alpha_2}(r), x_{\alpha_2+\alpha_3}(s)] = [x_{\alpha_1}(-r), x_{\alpha_2+\alpha_3}(s)][x_{\alpha_1+\alpha_2}(r), x_{\alpha_3}(s)].$$

By (vi), the last expression is the same as

$$[x_{\alpha_1+\alpha_2}(r), x_{\alpha_2+\alpha_3}(s)] = [x_{\alpha_1}(-r), x_{\alpha_2+\alpha_3}(s)][x_{\alpha_1}(r), x_{\alpha_2+\alpha_3}(s)].$$

Because by assumption there is an element $t \in R$ with $t, 1 - t \in R^\times$, it follows from (5.9) that $[x_{\alpha_1}(r), x_{\alpha_2+\alpha_3}(s)]$ is central in X , and hence in particular commutes with $x_{\alpha_1}(-r)$. Hence by (5.7),

$$[x_{\alpha_1+\alpha_2}(r), x_{\alpha_2+\alpha_3}(s)] = [x_{\alpha_1}(-r), x_{\alpha_2+\alpha_3}(s)][x_{\alpha_2+\alpha_3}(s), x_{\alpha_1}(-r)] = 1,$$

where the last equality is satisfied because $[x, y]^{-1} = [y, x]$ for any x, y in a group G . The relation (v) follows.

We now show (viii). Let $v_{123}(r)$ be as in (5.1). It suffices to show that $v_{123}(r)$ commutes with $x_{\alpha_2}(s)$ for all $r, s \in R$, since $v_{123}(r) = [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(r)]$ is central in X by (5.9), and X contains all the root subgroups associated to the roots $\{\alpha_1, \alpha_3, \alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_1 + \alpha_2 + \alpha_3\}$. But

$$\begin{aligned} v_{123}(r)^{x_{\alpha_2}(s)} &= [x_{\alpha_1+\alpha_2}(1)^{x_{\alpha_2}(s)}, x_{\alpha_3}(r)^{x_{\alpha_2}(s)}] \\ &= [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(r)x_{\alpha_2+\alpha_3}(-rs)] \\ &= [x_{\alpha_1+\alpha_2}(1), x_{\alpha_2+\alpha_3}(-rs)][x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(r)]^{x_{\alpha_2+\alpha_3}(-rs)} \\ &= [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(r)] = v_{123}(r). \end{aligned}$$

Here the second equality holds by (ii) and (iii), the third by (5.5) and the fourth by (v) and (5.9).

We now show (vii), with $x_{\alpha_1+\alpha_2+\alpha_3}(rs)$ replaced by $v_{123}(rs) = [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(rs)]$. Let $r, s \in R$. Then

$$\begin{aligned}
[x_{\alpha_1+\alpha_2}(r), x_{\alpha_3}(s)]^{x_{\alpha_2}(-r)} &= [[x_{\alpha_1}(1), x_{\alpha_2}(r)], x_{\alpha_3}(s)]^{x_{\alpha_2}(-r)} \\
&= [x_{\alpha_2}(-r), [x_{\alpha_3}(s), x_{\alpha_1}(-1)]]^{x_{\alpha_1}(1)} \cdot [x_{\alpha_1}(1), [x_{\alpha_2}(-r), x_{\alpha_3}(-s)]]^{x_{\alpha_3}(s)} \\
&= [x_{\alpha_1}(1), [x_{\alpha_2}(-r), x_{\alpha_3}(-s)]]^{x_{\alpha_3}(s)} \\
&= [x_{\alpha_1}(1), [x_{\alpha_2}(-r), x_{\alpha_3}(-s)]] \\
&= [x_{\alpha_1}(1), [x_{\alpha_2}(1), x_{\alpha_3}(rs)]] \\
&= [x_{\alpha_1}(1), x_{\alpha_2+\alpha_3}(rs)] = [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(rs)] = v_{123}(rs).
\end{aligned}$$

Here the first equality follows by (ii), the second by the Hall–Witt identity, the third by (iv), the fourth by (iv) and (iii), the fifth by (iii), and the last line by (vi). But by (viii),

$$[x_{\alpha_1+\alpha_2}(r), x_{\alpha_3}(s)] = v_{123}(rs)^{x_{\alpha_2}(r)} = v_{123}(rs),$$

and (vii) follows. Finally, for any $r, s \in R$,

$$\begin{aligned}
v_{123}(r+s) &= [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(r)x_{\alpha_3}(s)] \\
&= [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(s)][x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(r)]^{x_{\alpha_3}(s)} \\
&= [x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(s)][x_{\alpha_1+\alpha_2}(1), x_{\alpha_3}(r)] = v_{123}(s)v_{123}(r),
\end{aligned}$$

where the second equality follows from (5.5), and the third equality holds since $v_{123}(r)$ is central in X , which contains $x_{\alpha_3}(r)$. Hence (i) holds for $v_{123}(r)$, and this concludes the proof. $\square$

We are now able to prove Theorem A.

Corollary 5.8. *Let R be a (commutative, unital) ring such that 1_R is the sum of two invertible elements, and let $A = (A_{ij})_{i,j \in I}$ be spherical and simply laced. Then $U_R^+ \leq G_R$ is the amalgamated product of its subgroups U_{JR}^+ for $J \subseteq I$ and $|J| \leq 2$, along the canonical inclusions $U_{J_1R}^+ \leq U_{J_2R}^+$ for $J_1 \subseteq J_2 \subseteq I$.*

Proof. Since A is simply laced, it follows by Remark 5.2 that the groups U_{JR}^+ with $|J| = 2$ are generated by the root subgroups U_{α_j} for $j \in J$. Hence the statement follows immediately from Proposition 5.1 and Theorem 5.3. $\square$

5.2. On the condition that 1 is the sum of two units. A unital ring R is said to be *semi-local* if $R/J(R)$ is semisimple, with $J(R)$ its Jacobson radical. For commutative rings this is equivalent to having finitely many maximal ideals by the Chinese remainder theorem. The following result is classical, as it follows immediately from [Zel54] and [GPS98]. We include the proof for convenience of the reader.

Proposition 5.9. *Let R be a (not necessarily commutative) semi-local unital ring. Then the following are equivalent:*

- (i) R has no quotient isomorphic to $\mathbb{F}_2$,
- (ii) 1_R is the sum of two invertible elements,
- (iii) Every element of R is the sum of two invertible elements.

Proof. The implication (iii) $\implies$ (ii) is trivial. To see that (ii) implies (i), suppose $1_R = u + v$ with $u, v \in R^\times$. Suppose R has a quotient $\phi: R \twoheadrightarrow \mathbb{F}_2$. Then $\phi(u) = \phi(v) = 1$, since u and v are invertible and hence so is their image. But then $\phi(1_R) = \phi(u) + \phi(v) = 0$, a contradiction. Finally, to see that (i) implies (iii), let $J = J(R)$, the Jacobson radical of R . It is characterised as the ideal

$$J(R) = \{x \in R \mid 1 + yx \in R^\times, \text{ for all } y \in R\}.$$

First we claim that for $r \in R$, if $r + J$ is a sum of units in R/J , then r is a sum of units in R . It then suffices to show that every element of R/J is a sum of units.

First of all, if $a + J$ is invertible in R/J , then there is some $x \in R$ such that $xa + J = 1 + J$, and in particular, $xa - 1 \in J$, i.e. $xa \in R^\times$ by definition of J . It then follows that $a \in R^\times$.

Suppose now that $r + J$ is the sum of two units in R/J , i.e. $r + J = u + v + J$ for some $u + J, v + J \in (R/J)^\times$. Then there is some $j \in J$ such that $r = u + v + j$, and since u and v are invertible by the previous paragraph, we have that

$$r = u + v + j = u + v(1 + v^{-1}j).$$

Note that $1 + v^{-1}j \in R^\times$ since $j \in J$, and hence also $v(1 + v^{-1}j) \in R^\times$. In particular, r is the sum of two units, as desired.

We now show that every element in R/J is a sum of two units. Since R/J is semisimple, by the Wedderburn–Artin theorem R/J decomposes as

$$R/J \cong \prod_{i=1}^n M_{n_i}(D_i),$$

with each D_i a division ring. Since R is assumed to have no quotient $\mathbb{F}_2$, in particular no $M_{n_i}(D_i)$ in the above decomposition is isomorphic to $\mathbb{F}_2$. By [Zel54], not every element of a matrix ring $M_n(D)$ for D a division ring is a sum of two units if and only if $n = 1$ and $D = \mathbb{F}_2$. Hence for every i , every element $r \in M_{n_i}(D_i)$ is the sum of two units, and by [GPS98, Proposition 1.2], this property is preserved under taking direct products of rings. In particular, every element of R/J is a sum of two units, and the conclusion follows. $\square$

Remark 5.10. The equivalence of (i) and (iii) is studied more extensively in the literature. For example, the authors of [KS07] also show the result for right self-injective rings, using Kaplansky-style decompositions of such rings. The authors of *loc. cit.* also mention the existence of (non-commutative) von Neumann regular rings in which 2 is invertible (and hence 1 is a sum of two units, and the ring has no quotient $\mathbb{F}_2$) in which not every element is the sum of two units, attributed to Bergman (see [Han77]).

For our purposes we are mostly interested in the equivalence of (i) and (ii), since (i) is a necessary condition by Remark 5.5 and functoriality of U_R^+ in R , while (ii) is a sufficient condition. However, we were unable to find a result in the literature which completely classifies the rings R for which this equivalence holds, even when restricted to the case of commutative rings.

6. PRESENTATION OF $\overline{U}_R^+$

In this section, we now prove Corollary C and Corollary D. We first state the following result, due to Stein ([Ste73]). See also [MM25, Theorem 5.5].

Lemma 6.1 ([Ste73]). *Let R be a local domain and let A be spherical. Then $\varphi_R: G_R \rightarrow G_R^{\min}$ is a bijection. In particular, $U_R^\pm \cong \overline{U}_R^\pm$.*

The following theorem shows the reverse implication in Corollary C.

Theorem 6.2. *Suppose A is a 3-spherical simply laced GCM, and let R be a local Bézout domain with residue field different from $\mathbb{F}_2$. Then $\overline{U}_R^+$ is the amalgamated product of its subgroups $\overline{U}_{JR}^+$ with $J \subseteq I$ and $|J| \leq 2$, along the inclusions $\overline{U}_{J_1R}^+ \leq \overline{U}_{J_2R}^+$ for all $J_1 \subseteq J_2$.*

Proof. By Theorem 4.11, it suffices to show the statement for all rank 3 GCM's A of simply laced spherical type. In this case, $\overline{U}_R^+ \cong U_R^+$ by Lemma 6.1. We distinguish three cases for A .

If A is indecomposable, then it is of type A_3 , and the statement follows from Theorem 5.3, since by Proposition 5.9, 1_R is a sum of two units.

If A is decomposable, then $U_R^+ \cong U_{A'R}^+ \times (R, +)$, where A' is either of type A_2 or of type $A_1 \times A_1$, i.e.

$$U_R^+ \cong \left\{ \begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix} \in \mathrm{SL}_3(R) \right\} \times (R, +) \quad \text{or} \quad U_R^+ \cong (R, +) \times (R, +) \times (R, +).$$

It is then immediate from the relations (2.1) that U_R^+ has the desired presentation. $\square$

We now prove Corollary D. In [MM25, Theorem A], this result was already proven in the more general setting of arbitrary 2-spherical GCM's, but as a consequence of our work we obtain an alternative proof in the 3-spherical simply laced case.

Proposition 6.3. *Let $A = (A_{ij})_{i,j \in I}$ be 3-spherical and simply laced. Let R be a local Bézout domain with residue field different from $\mathbb{F}_2$, and with field of fractions $\mathbb{K}$. Then the map*

$$\varphi_R: G_R \rightarrow G_R^{\min} \subseteq G_{\mathbb{K}}$$

is injective.

Proof. By Theorem 6.2, $\overline{U}_R^+$ is the free product of the root subgroups $U_{\alpha R}$, for the roots $\alpha \in \Psi$, where

$$\Psi := \{\alpha \in \Phi \mid \exists i, j \in I \text{ such that } \alpha \in (\mathbb{N}\alpha_i + \mathbb{N}\alpha_j) \cap \Phi^+\},$$

subject to the relations (2.1) for all pairs $\{\alpha, \beta\} \subseteq \Psi$ with $\alpha \neq \pm\beta$. But these relations are all already satisfied in U_R^+ , and since moreover U_R^+ is generated by all the simple root subgroups $U_{\alpha_i R}$ by Lemma 2.3, it follows that $\varphi_R|_{U_R^+}$ is injective.

The proof of [MM25, Theorem 5.5], now applies *verbatim*. $\square$

We now show the forward implication in Corollary C.

Proposition 6.4. *Suppose $A = (A_{ij})_{i,j \in I}$ is simply laced and R is a local Bézout domain with residue field different from $\mathbb{F}_2$. If $\overline{U}_R^+$ is isomorphic to the amalgamated product of its subgroups $\overline{U}_{J_R}^+$ with $J \subseteq I$ and $|J| \leq 2$ along the inclusions $\overline{U}_{J_1 R}^+ \leq \overline{U}_{J_2 R}^+$ for $J_1 \subseteq J_2$, then A is 3-spherical.*

Proof. By Proposition 6.3, φ_R is injective, and hence $\overline{U}_R^+$ is isomorphic to $U_R^+ \leq G_R$. The result then immediately follows from [CM26, Theorem E]. $\square$

Corollary C now follows from Theorem 6.2 and Proposition 6.4.

Finally, note that the stabilisers in $\overline{B}_R^+$ of the J -residues $\overline{U}_R^+ c_0^- = \overline{B}_R^+ c_0^- \subseteq C^-(G_R^{\min})$ are given by

$$\overline{B}_R^+ \cap \overline{P}_{J_R}^- = (\overline{U}_R^+ \cap \overline{P}_{J_R}^-) T_R = \overline{U}_{J_R}^+ T_R,$$

where the last equality follows by (2.9). We then deduce the following.

Corollary 6.5. *Let R be a local Bézout domain with residue field different from $\mathbb{F}_2$. Let A be a simply laced GCM. Then $\overline{B}_R^+$ is the amalgamated product of its subgroups $\overline{U}_{J_R}^+ T_R$ for all $J \subseteq I$ with $|J| \leq 2$, along the inclusions $\overline{U}_{J_1 R}^+ T_R \leq \overline{U}_{J_2 R}^+ T_R$ for $J_1 \subseteq J_2$, if and only if A is 3-spherical.*

Note also that applying the Cartan–Chevalley involution ω_R , the analogue of any result stated in this paper for $\overline{U}_R^+$ and U_R^+ also holds for $\overline{U}_R^-$ and U_R^- .

REFERENCES

- [AB08] Peter Abramenko and Kenneth S. Brown. *Buildings*, volume 248 of *Graduate Texts in Mathematics*. Springer, New York, 2008. Theory and applications.
- [Abr96] Peter Abramenko. *Twin buildings and applications to S-arithmetic groups*, volume 1641 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, 1996.
- [All16] Daniel Allcock. Steinberg groups as amalgams. *Algebra Number Theory*, 10(8):1791–1843, 2016.
- [CM26] Pierre-Emmanuel Caprace and Timothée Marquis. Amalgams of rational unipotent groups and residual nilpotence. *J. Comb. Algebra*, 2026. Published online first.
- [DM07] Alice Devillers and Bernhard Mühlherr. On the simple connectedness of certain subsets of buildings. *Forum Math.*, 19(6):955–970, 2007.
- [GPS98] B. Goldsmith, S. Pabst, and A. Scott. Unit sum numbers of rings and modules. *Quart. J. Math. Oxford Ser. (2)*, 49(195):331–344, 1998.
- [Han77] David Handelman. Perspectivity and cancellation in regular rings. *J. Algebra*, 48(1):1–16, 1977.

- [HNVM16] Max Horn, Reed Nessler, and Hendrik Van Maldeghem. Simple connectivity in polar spaces. *Forum Math.*, 28(3):491–505, 2016.
- [Hum90] James E. Humphreys. *Reflection groups and Coxeter groups*, volume 29 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 1990.
- [KS07] Dinesh Khurana and Ashish K. Srivastava. Right self-injective rings in which every element is a sum of two units. *J. Algebra Appl.*, 6(2):281–286, 2007.
- [Mar18] Timothée Marquis. *An introduction to Kac–Moody groups over fields*. EMS Textbooks in Mathematics. European Mathematical Society (EMS), Zürich, 2018.
- [MM25] Timothée Marquis and Bernhard Mühlherr. Presentation and uniqueness of Kac–Moody groups over local rings. Preprint, 2025. arXiv:2510.11272.
- [Ron09] Mark Ronan. *Lectures on buildings*. University of Chicago Press, Chicago, IL, 2009. Updated and revised.
- [Ser80] Jean-Pierre Serre. *Trees*. Springer-Verlag, Berlin-New York, 1980. Translated from the French by John Stillwell.
- [Ste73] Michael R. Stein. Surjective stability in dimension 0 for K_2 and related functors. *Trans. Amer. Math. Soc.*, 178:165–191, 1973.
- [Tit86] Jacques Tits. Ensembles ordonnés, immeubles et sommes amalgamées. *Bull. Soc. Math. Belg. Sér. A*, 38:367–387, 1986.
- [Tit87] Jacques Tits. Uniqueness and presentation of Kac-Moody groups over fields. *J. Algebra*, 105(2):542–573, 1987.
- [Zel54] Daniel Zelinsky. Every linear transformation is a sum of nonsingular ones. *Proc. Amer. Math. Soc.*, 5:627–630, 1954.

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